

Chapter 3. Chemical Kinetics

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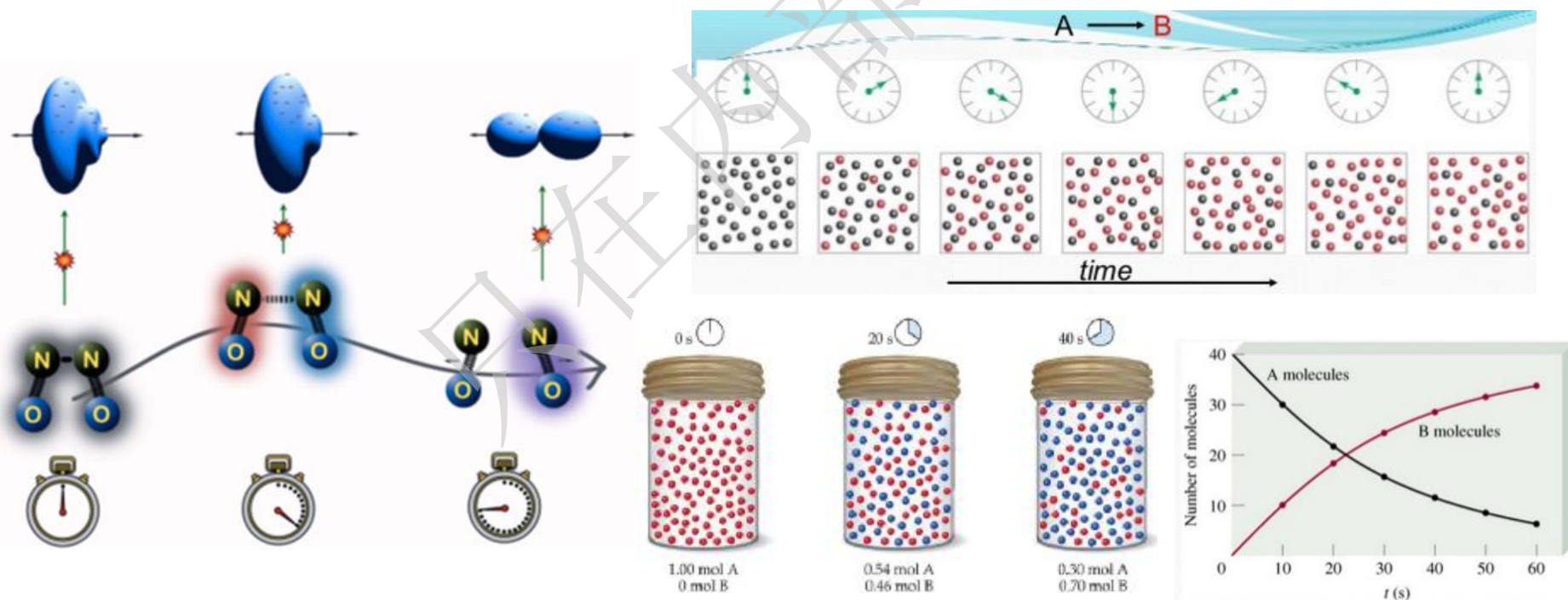
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Outlines

- 1. Fundamentals of Chemical Kinetics**
- 2. Bimolecular Reaction Theory**
- 3. Unimolecular Reaction Theory**
- 4. Recommended Literature**

1. Fundamentals of Chemical Kinetics

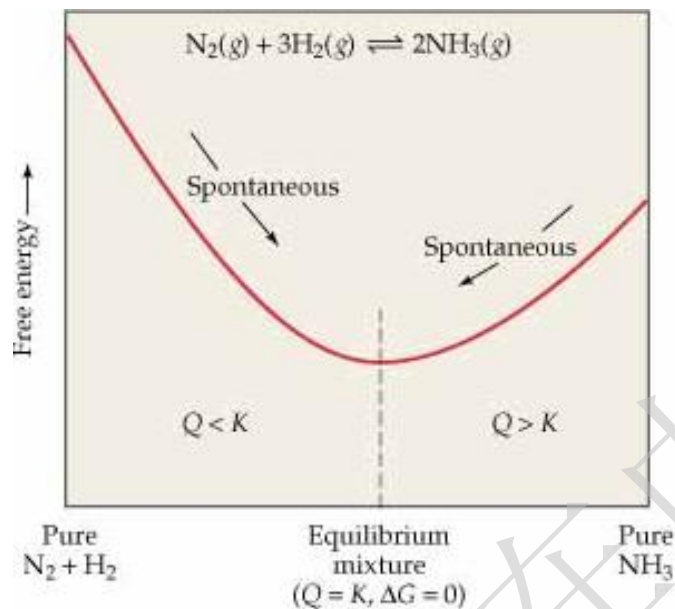
- Study the rates of chemical processes
- Describe **how fast** the reaction proceeds
- Determine **whether** under a particular reaction condition, the reaction can proceed to the equilibrium state
- Study the factors which affect the reaction rates



1. Fundamentals of Chemical Kinetics

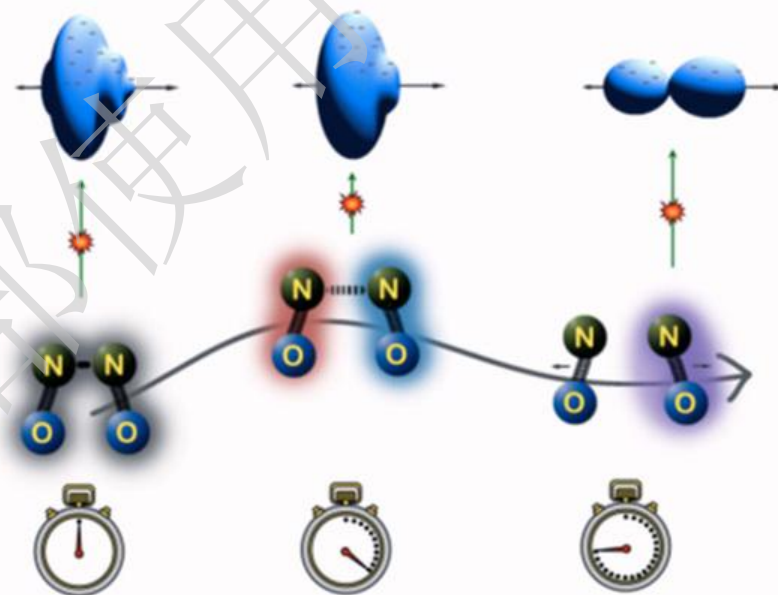
I Difference between Thermodynamics and Kinetics

Thermodynamics



- n Reaction processes: spontaneous, non-spontaneous, equilibrium
- n Determine **whether** the reaction process is a spontaneous process
- n Determine the **maximum extend** (equilibrium) the spontaneous reaction processes

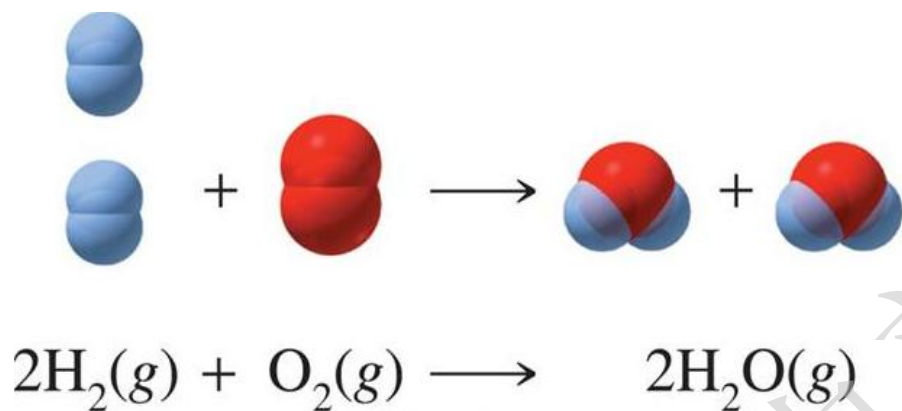
Kinetics



- n Determine how fast the reaction proceeds
- n Determines whether under a particular condition, the reaction can proceed to the equilibrium state

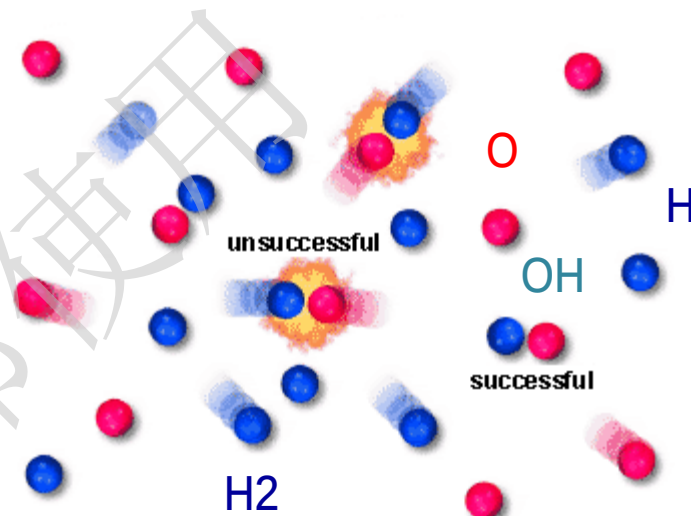
1. Fundamentals of Chemical Kinetics

I Global & Elementary Reactions



Global reaction

- n Describe an overall process
- n Do not describe the actual physical process
- n Physically impossible (collision)
- n Statistically is a unlikely event



Elementary reaction

- n Take place physically by collision
- n All elementary reactions are reversible

1. Fundamentals of Chemical Kinetics

I Elementary Reaction

Reaction Type, Reaction Rates, Reaction order



Unimolecular reaction

n Examples

p Bond dissociation reactions

p Isomerization reactions

n First order reaction $k(\text{s}^{-1})$

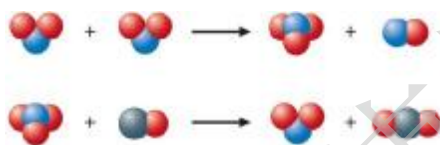


$$\frac{-d[A]}{dt} = k [A]$$

n Second order reaction



$$\frac{-d[A]}{dt} = k' [A][M]$$



Bimolecular reaction

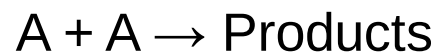
n Second order reaction

$$k(\text{cm}^3/\text{mol}\cdot\text{s})$$

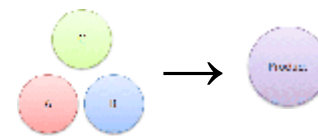


$$\frac{-d[A]}{dt} = \frac{-d[B]}{dt} = k [A][B]$$

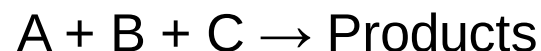
n Pseudo-first order reaction



$$-\frac{1}{2} \frac{d[A]}{dt} = k [A]^2$$



Termolecular reaction



n Very rare

n Require the simultaneous collision of three molecules

1. Fundamentals of Chemical Kinetics

I Law of Mass Action

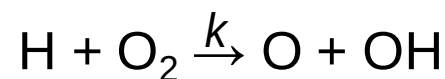
- n The rate of a chemical reaction is proportional to the product of the **masses** (concentrations) of the reactants

$$r_i = k_i \prod_j^{N_s} Y_j^{\nu_{ij}^L}$$

Annotations for the equation:

- r_i : Rate($d[i]/dt$)
- k_i : Rate coefficient
- ν_{ij}^L : Stoichiometric coefficient
- Y_j : Concentration of species j

Example



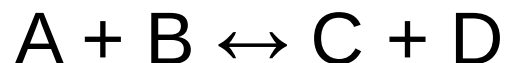
$$r_{\text{OH}} = k \cdot [\text{H}] \cdot [\text{O}_2]$$

- n The **kinetic** aspect: concerning **rate equations** for elementary reactions
- n The **equilibrium** aspect: 1) chemical equilibrium is a dynamic process in which rates of reaction for the forward and backward reactions must be equal at chemical equilibrium; 2) a chemical equilibrium can be shifted towards the products by adding reactants and towards the reactants by adding products to the reacting mixture

**What
principle
is it?**

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I Reversible Reaction



$$R_{forward} = R_f = k_f [A][B]$$

$$R_{reverse} = R_b = k_b [C][D]$$

Net reaction rate:

$$\begin{aligned} -\frac{d[A]}{dt} &= -\frac{d[B]}{dt} = \frac{d[C]}{dt} = \frac{d[D]}{dt} \\ &= R_f - R_b = k_f [A][B] - k_b [C][D] \end{aligned}$$

n When the reaction reaches chemical equilibrium, **the net rate goes to zero**

n The forward and reverse reactions proceed at **finite and equal rates**

$$R_f - R_b = k_f [A]_{eq}[B]_{eq} - k_b [C]_{eq}[D]_{eq} = 0$$

$$\frac{k_f}{k_b} = \frac{[A]_{eq}[B]_{eq}}{[C]_{eq}[D]_{eq}}$$

n Equilibrium constant K_c

$$K_c = \frac{k_f}{k_b}$$

n The forward and reverse reaction rate constants are related to the equilibrium constant, do not have to be independently determined if the thermodynamic data of reactants and products or the equilibrium constant are known

1. Fundamentals of Chemical Kinetics

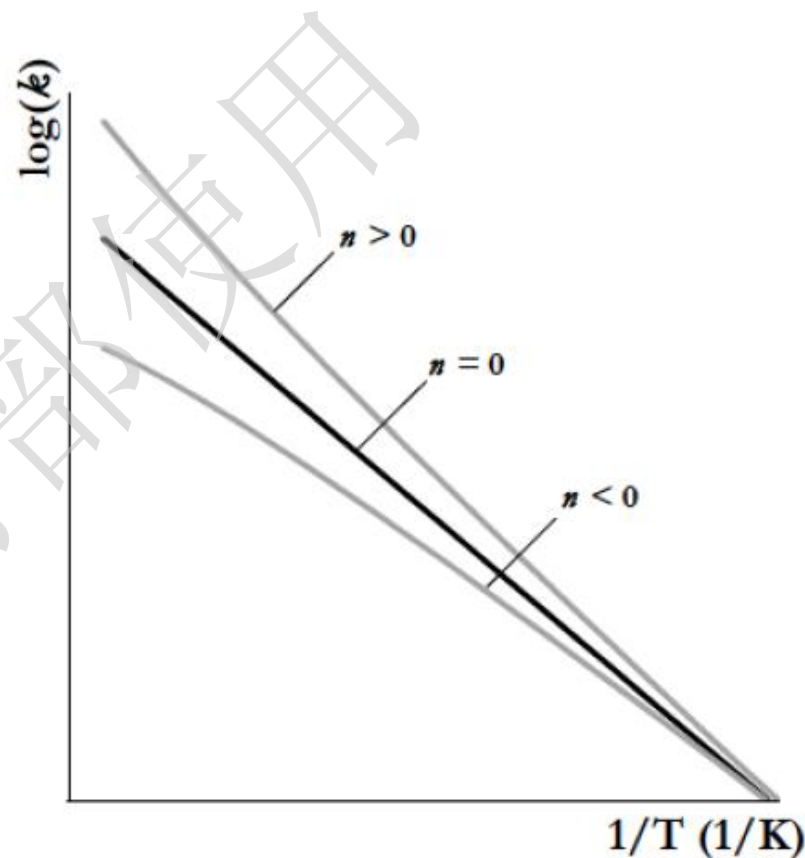
I Arrhenius Law

- n Rate constants of **most** reactions are strongly dependent on temperature. This dependency is given by the Arrhenius Law as:

$$k = A \exp\left(-\frac{E_a}{RT}\right)$$

- n In general, the pre-factor A is not a constant. A modified Arrhenius expression may be introduced to account for this effect:

$$k = A T^n \exp\left(-\frac{E_a}{RT}\right)$$



An Arrhenius plot

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I Activation Energy

- n The activation energy represents the **potential energy level** in a molecule above which the reaction is possible
- n The origin of the activation energy is the Maxwell-Boltzmann distribution of energy:

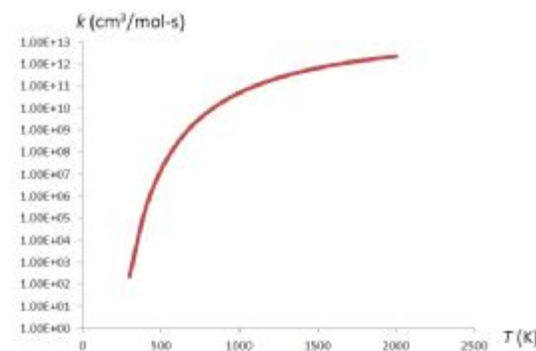
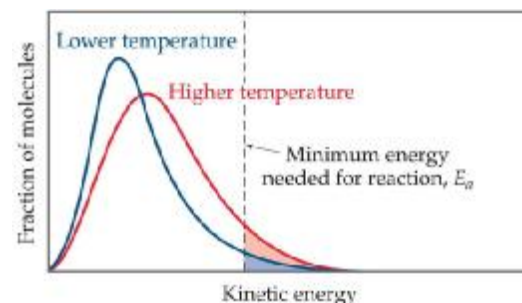
$$B(E) = \frac{N_E}{N_{tot}} = \frac{1}{Q} \exp\left(-\frac{E}{k_B T}\right)$$

- n As T increases, rate constant curve tends to be flat at high temperatures

Is the Arrhenius Law always the truth?

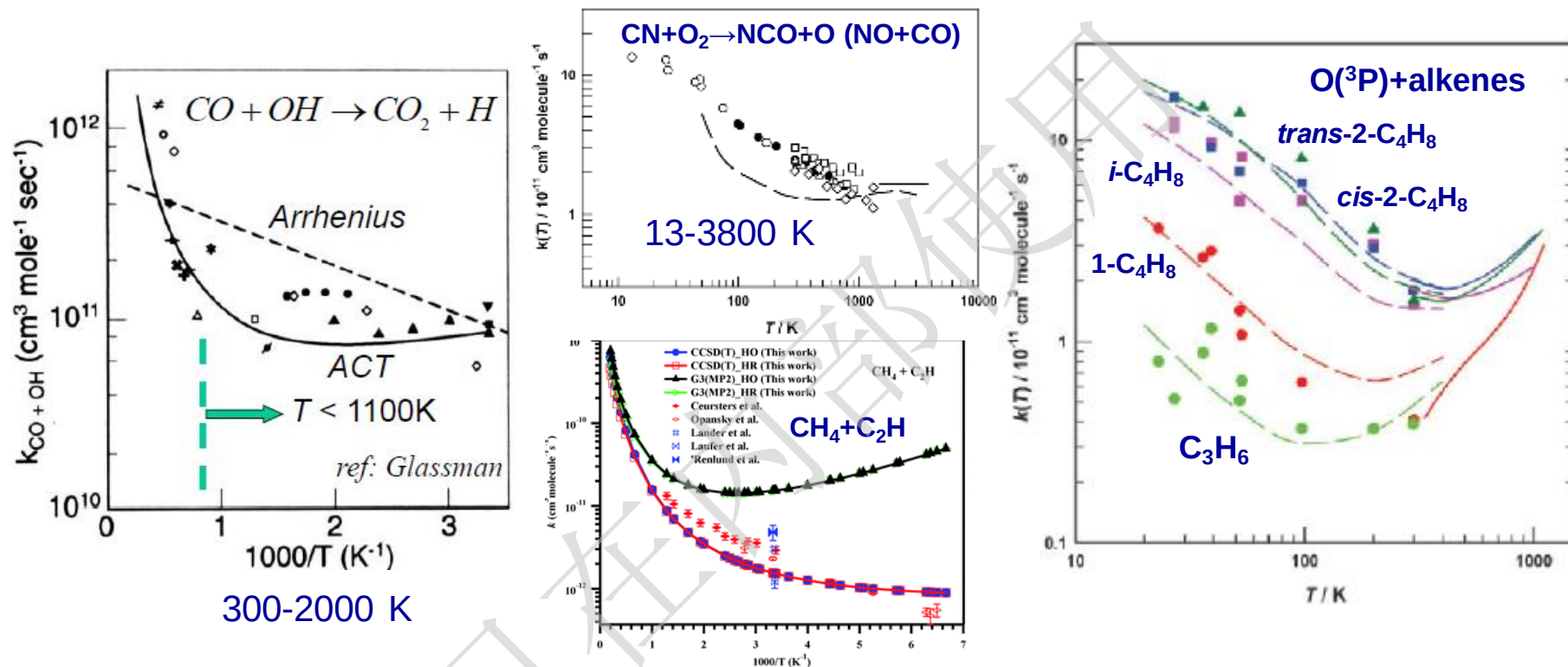


Maxwell-Boltzmann Distributions



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I Non-Arrhenius Behavior



n For the **wide temperature ranges** found in combustion, “non-Arrhenius” behavior of bimolecular reaction rate constants is common, the behavior is **not exception**

Why?

1. Fundamentals of Chemical Kinetics

I Complex Reaction System

- n Most real chemical processes are complex reaction systems consisting of a lot of elementary reactions
- n Reaction mechanism – List of reactions + corresponding rate coefficients
- n Large, complicated reaction mechanism $\Rightarrow$ large, complicated **kinetic system of differential equations**
- n Simplification principles – can be used for the simplification of a reaction mechanism in such a way that the obtained reaction mechanism provides an **almost identical** solution
- n Commonly used kinetics simplification principles
 - p Quasi steady state approximation (QSSA)
 - p Rate determining step
 - p Fast pre-equilibrium approximation

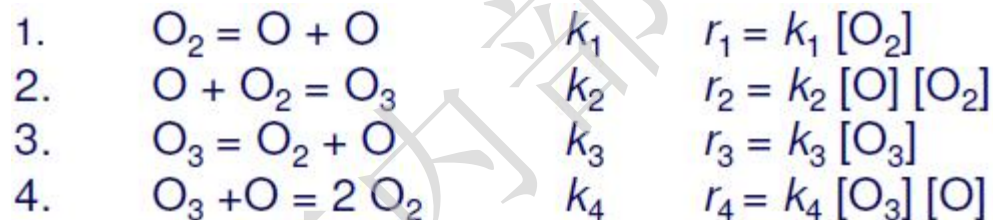
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I Kinetic system of differential equations

- n For complex reaction system, many reactions occur simultaneously, called kinetic system of differential equations (equation set)

$$\frac{dY_j}{dt} = \sum_i \nu_{ij} r_i; \quad j = 1, 2, \dots, n$$

- n Example: Chapman mechanism



- n The change of concentrations in time for each species can be calculated via the solution of the kinetic system of differential equations:

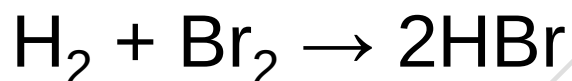
$$\frac{d[O_2]}{dt} = -1r_1 - 1r_2 + 1r_3 + 2r_4 \quad \frac{d[O_2]}{dt} = -1k_1[O_2] - 1k_2[O][O_2] + 1k_3[O_3] + 2k_4[O][O_3]$$

$$\frac{d[O]}{dt} = 2r_1 - 1r_2 + 1r_3 - 1r_4 \quad \frac{d[O]}{dt} = 2k_1[O_2] - 1k_2[O][O_2] + 1k_3[O_3] - 1k_4[O][O_3]$$

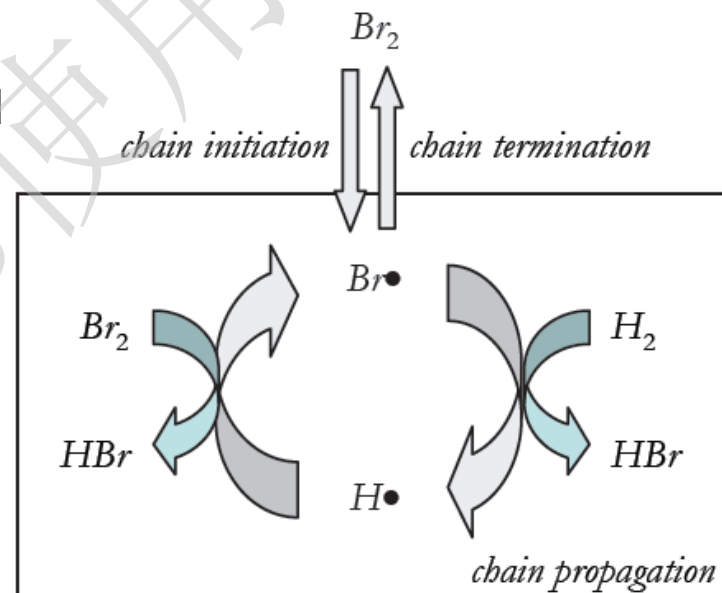
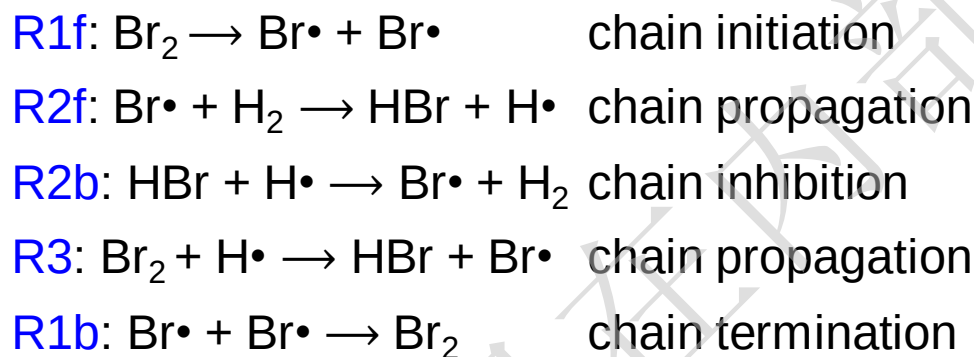
$$\frac{d[O_3]}{dt} = 1r_2 - 1r_3 - 1r_4 \quad \frac{d[O_3]}{dt} = 1k_2[O][O_2] - 1k_3[O_3] - 1k_4[O][O_3]$$

1. Fundamentals of Chemical Kinetics

I Chain Reactions



- n Experimental observation showed $d[\text{HBr}]/dt \sim ([\text{Br}_2])^{1/2}$ at an early reaction time, which could not be explained using global mechanism



- n The production rates of the **chain carriers**

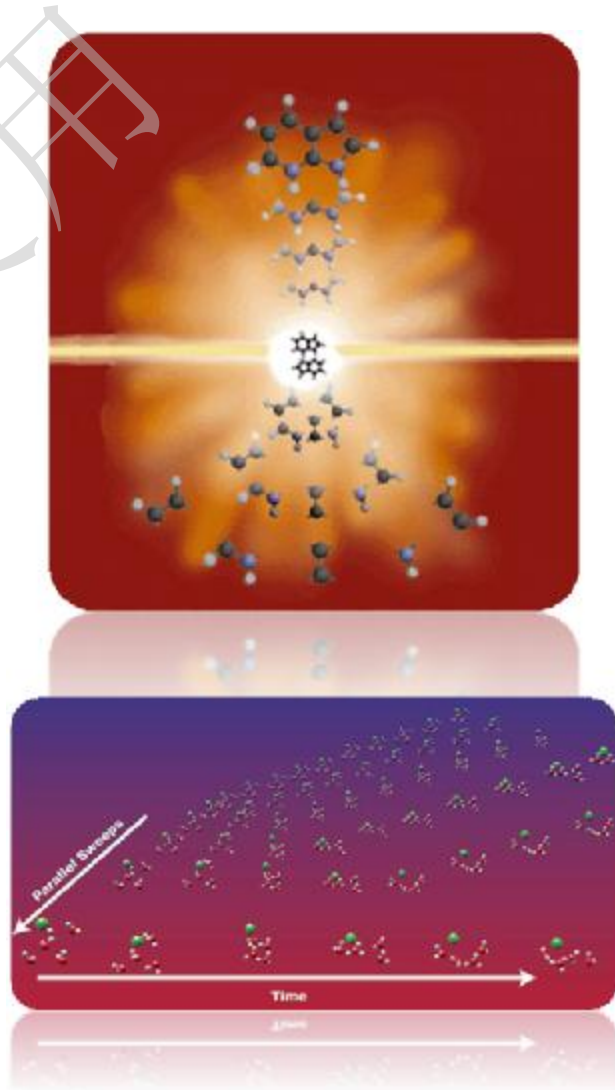
$$\frac{d[\text{Br}\cdot]}{dt} = 2k_{1f} [\text{Br}_2] - k_{2f} [\text{Br}\cdot][\text{H}_2] + k_{2b} [\text{HBr}][\text{H}\cdot] + k_3 [\text{H}\cdot][\text{Br}_2] - 2k_{1b} [\text{Br}\cdot]^2$$

$$\frac{d[\text{H}\cdot]}{dt} = k_{2f} [\text{Br}\cdot][\text{H}_2] - k_{2b} [\text{HBr}][\text{H}\cdot] - k_3 [\text{H}\cdot][\text{Br}_2]$$

1. Fundamentals of Chemical Kinetics

I Chain Reactions

- n In the **chain initiation** steps, chain carriers (also called chain centres, i.e. reactive intermediates) are generated.
- n In the **chain propagation** steps, the chain carriers react with the reactants, produce products and regenerate the chain carriers.
- n In the **chain inhibition** step, the chain carriers react with the product, reactants are reformed, and there is no reduction in the number of chain carriers.
- n In the **chain branching** step two or more chain carriers are produced from a single chain carrier.
- n In the **chain termination** steps the chain carriers are consumed.

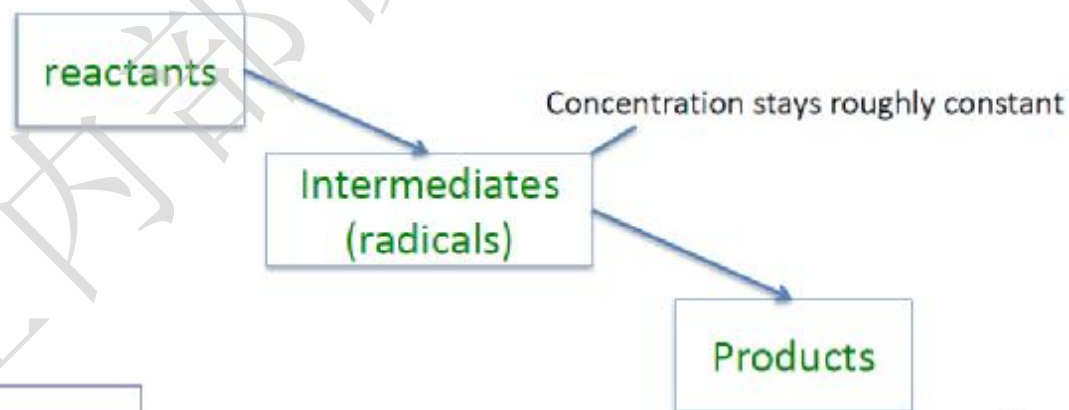


1. Fundamentals of Chemical Kinetics

I Quasi-Steady-State Assumption

- Highly **reactive intermediate species** such as radicals are usually present only in small concentrations
- They do not accumulate during the reaction process, the production rate is low while the consumption rate is high $\frac{d[\text{free radical}]}{dt} \leq \frac{d[\text{reactant}]}{dt}$
- QSSA: $\frac{d[\text{free radical}]}{dt} \sim 0$

NOTE: QSSA is a useful tool in reaction analysis and mechanism reduction since it can simplify ODE equations to algebraic equations



Scheme I:		$A \xrightarrow{k_1} B \xrightarrow{k_2} C$		(35)
Exact Solution		QSSA Solution		
$c_A = c_{A0}e^{-k_1 t}$	(36)	$c_{As} = c_{A0}e^{-k_1 t}$	(39)	
$c_B = c_{A0} \frac{k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$	(37)	$c_{Bs} = c_{A0} \frac{k_1}{k_2} e^{-k_1 t}$	(40)	
$c_C = c_{A0} \frac{1}{k_2 - k_1} (k_2(1 - e^{-k_1 t}) - k_1(1 - e^{-k_2 t}))$	(38)	$c_{Cs} = c_{A0} (1 - e^{-k_1 t})$	(41)	

After-class thinking

A lot of intermediates can be treated with QSSA, does it mean all of them can be ignored in real combustion systems?

1. Fundamentals of Chemical Kinetics

I QSSA of Chain Reactions

n Use QSSA to treat $\text{Br}\cdot$ and $\text{H}\cdot$

$$\frac{d[\text{Br}\cdot]_{qss}}{dt} = 2k_{1f}[\text{Br}_2] - k_{2f}[\text{Br}\cdot]_{qss}[\text{H}_2] + k_{2b}[\text{HBr}][\text{H}\cdot]_{qss} + k_3[\text{H}\cdot]_{qss}[\text{Br}_2] - 2k_{1b}[\text{Br}\cdot]_{qss}^2 = 0$$

$$\frac{d[\text{H}\cdot]_{qss}}{dt} = k_{2f}[\text{Br}\cdot]_{qss}[\text{H}_2] - k_{2b}[\text{HBr}][\text{H}\cdot]_{qss} - k_3[\text{H}\cdot]_{qss}[\text{Br}_2] = 0$$

n Obtain

$$[\text{Br}\cdot]_{qss} = \sqrt{\frac{k_{1f}[\text{Br}_2]}{k_{1b}}} \quad [\text{H}\cdot]_{qss} = \sqrt{\frac{k_{1f}[\text{Br}_2]}{k_{1b}}} \frac{k_{2f}[\text{H}_2]}{k_{2b}[\text{HBr}] + k_3[\text{Br}_2]}$$

n The production rate of HBr

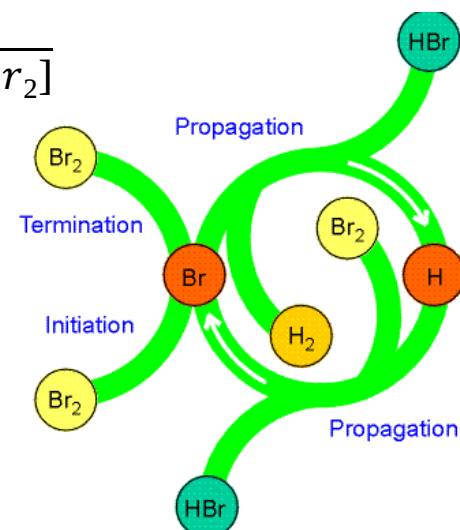
$$\frac{d[\text{HBr}]}{dt} = k_{2f}[\text{Br}\cdot]_{qss}[\text{H}_2] - k_{2b}[\text{HBr}][\text{H}\cdot]_{qss} + k_3[\text{H}\cdot]_{qss}[\text{Br}_2]$$

$$\frac{d[\text{HBr}]}{dt} = k_{2f} \sqrt{\frac{k_{1f}[\text{Br}_2]}{k_{1b}}} [\text{H}_2] \left(1 + \frac{k_{2b}[\text{HBr}]}{k_3[\text{Br}_2]}\right)^{-1}$$

n At low reaction conversion (early reaction time), $[\text{HBr}] \sim 0$

$$\frac{d[\text{HBr}]}{dt} = k_{2f} \sqrt{\frac{k_{1f}[\text{Br}_2]}{k_{1b}}} [\text{H}_2]$$

n Explains the experimental observation for the $([\text{Br}_2])^{1/2}$ dependence



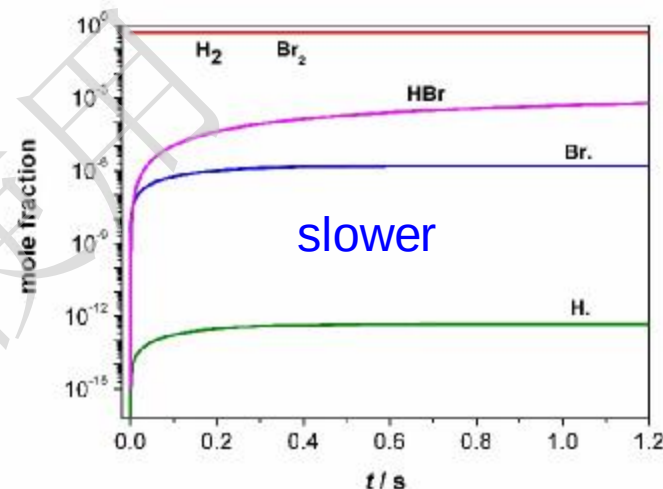
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I Basic Types of Chain Reactions

The two basic types of chain reactions

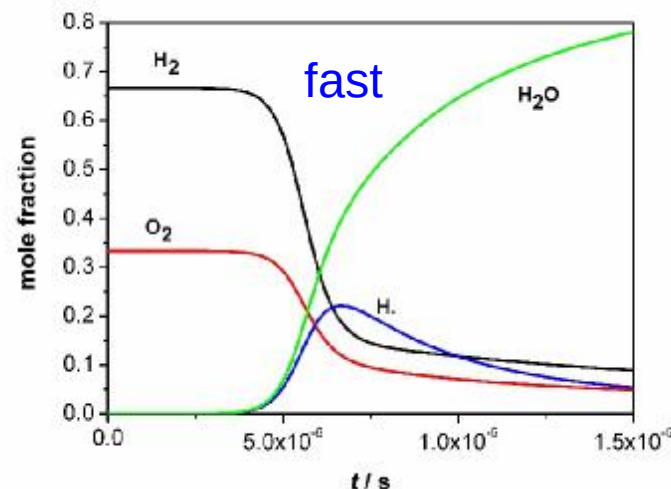
Open Chain Reactions

- n Chain reactions without branching steps
- n Examples: $\text{H}_2 + \text{Br}_2$ reaction, alkane pyrolysis and polymerization reactions



Branched chain reactions

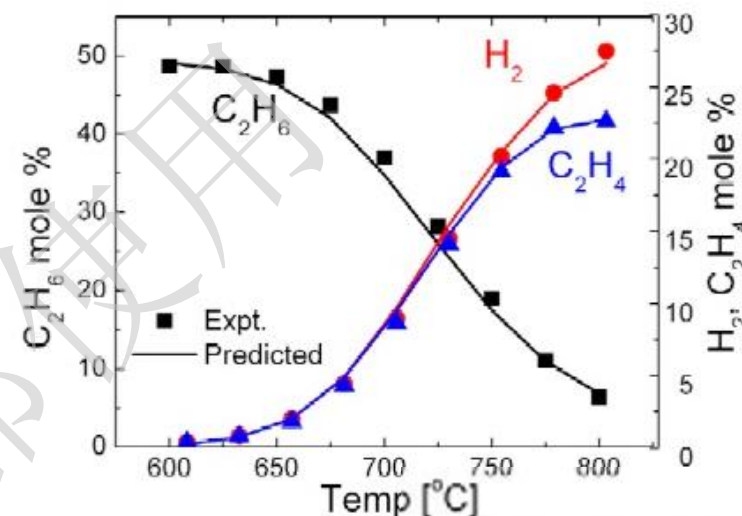
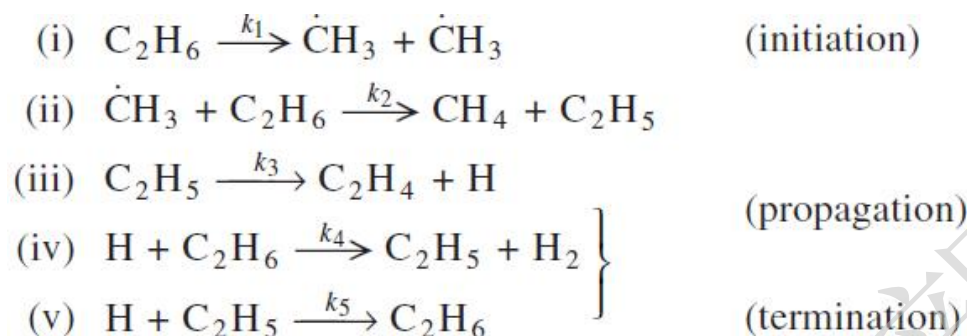
- n Chain reactions that include branching reaction steps
- n Examples: $\text{H}_2 + \text{O}_2$ reaction, hydrocarbon/air explosions and flames
- n Fast reactions and explosion



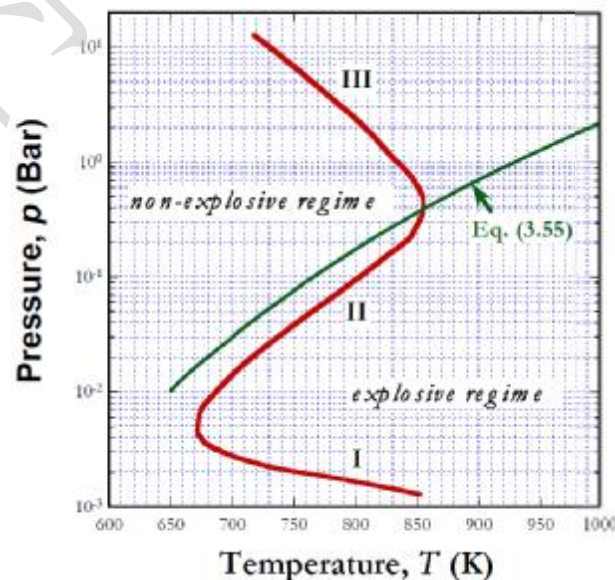
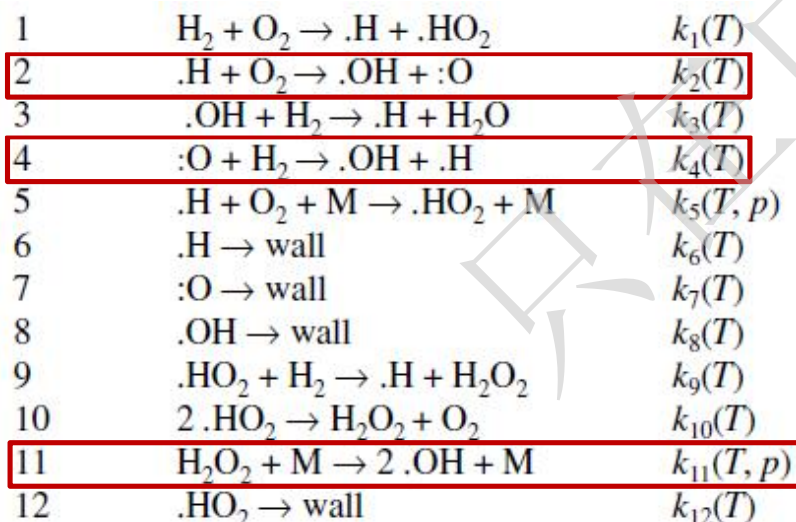
1. Fundamentals of Chemical Kinetics

I Basic Types of Chain Reactions

Thermal decomposition of ethane



Branching Chain Explosion



1. Fundamentals of Chemical Kinetics

I Partial-Equilibrium Approximation

- n Applied to specific **reaction(s)** (not specific species)
- n A species participates in a pair of **fast-equilibrium** reactions and then is consumed by slow reaction(s)
- n The fast reactions are rapid in both the forward and reverse directions
- n Treating fast reactions **as if they are equilibrated**, the rates of the forward and backward reactions become equal to each other

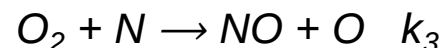
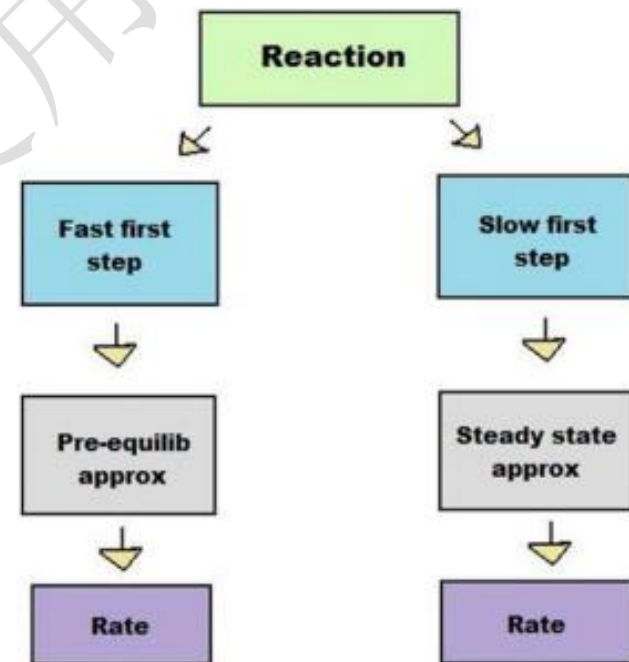
$$\frac{d[N]}{dt} = k_2[N_2][O] - k_3[O_2][N] \approx 0 \quad (\text{QSSA})$$

$$\frac{d[NO]}{dt} = k_2[N_2][O] + k_3[O_2][N] = 2k_2[N_2][O]$$

$$k_{1f}[O_2] = k_{1b}[O]^2 \quad (\text{PEA}) \quad [O] = \sqrt{K_c[O_2]} \quad K_c = k_{1f}/k_{1b}$$

$$\frac{d[NO]}{dt} = 2k_2\sqrt{K_c}[N_2][O_2]^{1/2}$$

PEA & QSSA

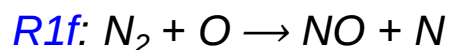


1. Fundamentals of Chemical Kinetics

I Rate-Determining Step

- For a complex reaction, the consumption rate of a reactant or the production rate of a final product may depend mainly on the rate coefficient of a single reaction step. This reaction step is called the rate-determining (or rate-limiting) step

Zel'dovich Mechanism



$$k_{1f} = 1.8 \times 10^{14} e^{-38370K/T}$$



$$k_{1b} = 3.8 \times 10^{13} e^{-425K/T}$$



$$k_{2f} = 1.8 \times 10^{10} T e^{-4680K/T}$$



$$k_{2b} = 3.8 \times 10^9 T e^{-20820K/T}$$

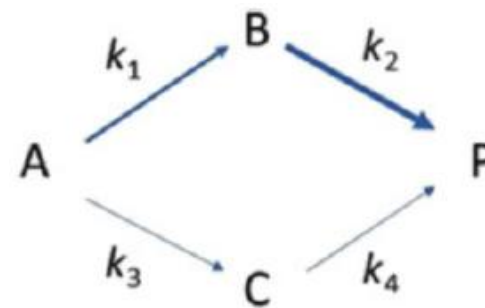
Characteristic Reaction Times $\tau_{chem} \equiv \frac{[N] - [N]_0}{d[N]/dt}$

	1000 K	1500 K	2000 K
τ_{1f}	3×10^7 s	100 s	0.2 s
τ_{2f}	2 μ s	0.5 μ s	0.2 μ s
τ_{1r}	300 μ s	400 μ s	500 μ s

- Overall NO production is limited by the **slow** reaction R_{1f}

- In the case of an arbitrary mechanism, the rate-limiting step **may not be** the reaction step having the smallest rate coefficient (**Competition**)

- Sensitivity analysis** was performed to determine the rate-limiting step (**The role of reaction flux**)



If $k_3, k_4 \leq k_1 \leq k_2$, k_1 is the rate-limiting step

1. Fundamentals of Chemical Kinetics

I Local Sensitivity Analysis

- n For a complex reaction system, assume that the solution of the system of differential equations is calculated from time t_0 to time t_1

$$\frac{d\mathbf{Y}}{dt} = f(\mathbf{Y}, \mathbf{x}), \quad Y(t_0) = Y_0$$

- n The value of parameter x_j is changed by Δx_j , the solution of the ODE is continued until time t_2 , yield solution $\tilde{Y}_i(t_2)$. The sensitivity coefficient :

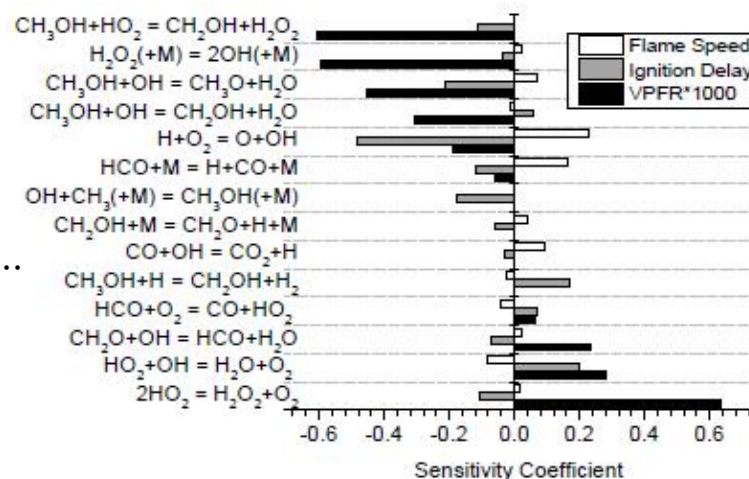
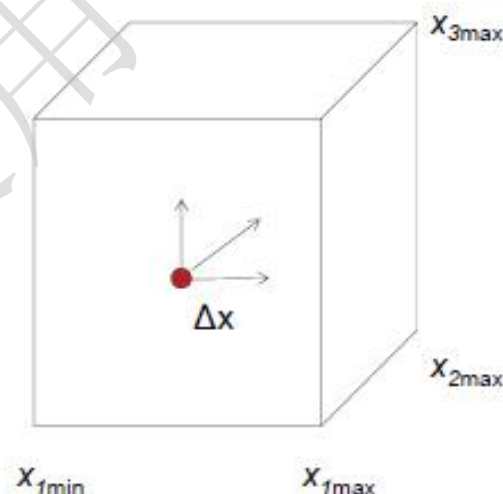
$$\frac{\partial Y_i}{\partial x_j}(t_1, t_2) \approx \frac{\Delta Y_i(t_2)}{\Delta x_j} = \frac{\tilde{Y}_i(t_2) - Y_i(t_2)}{\Delta x_j}$$

- n Effect on the model solutions: **Second order**

$$Y_i(t, \mathbf{x} + \Delta \mathbf{x}) = Y_i(t, \mathbf{x}) + \sum_{j=1}^m \frac{\partial Y_i}{\partial x_j} \Delta x_j + \frac{1}{2} \sum_{k=1}^m \sum_{j=1}^m \frac{\partial^2 Y_i}{\partial x_k \partial x_j} \Delta x_k \Delta x_j + \dots$$

- n Normalized sensitivity coefficients: $\frac{x_j}{y_i} \frac{\partial Y_i}{\partial x_j}$

Local @ nominal values



1. Fundamentals of Chemical Kinetics

I Local Sensitivity Analysis

Local sensitivity analysis features

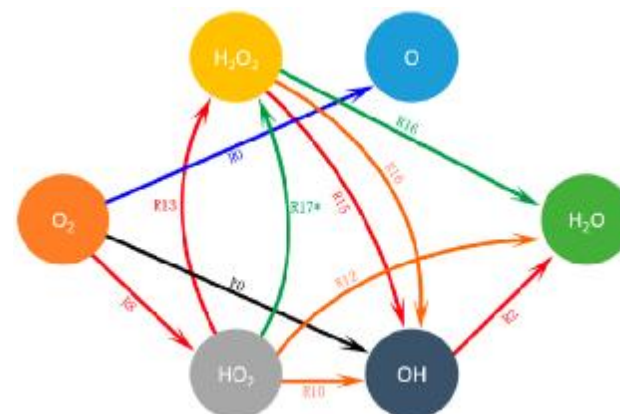
- n** If a rate coefficient is changed, it directly influences the concentrations of those species that are present in the corresponding chemical reaction step
- n** These changes influence the concentrations of other species through coupled reaction steps
- n** For any other model which is based on differential equations using a similar approach, both direct and indirect effects of parameter changes could be identified

p Local sensitivity matrix

p Depend on the time t_1 of the parameter perturbation and the time t_2 of inspection of the effect of the parameter perturbation

$$\frac{d}{dt} \frac{\partial Y}{\partial x_j} = J \frac{\partial Y}{\partial x_j} + \frac{\partial f}{\partial x_j}, \quad \frac{\partial Y}{\partial x_j}(t_0) = 0$$

$(j = 1, 2 \dots N)$



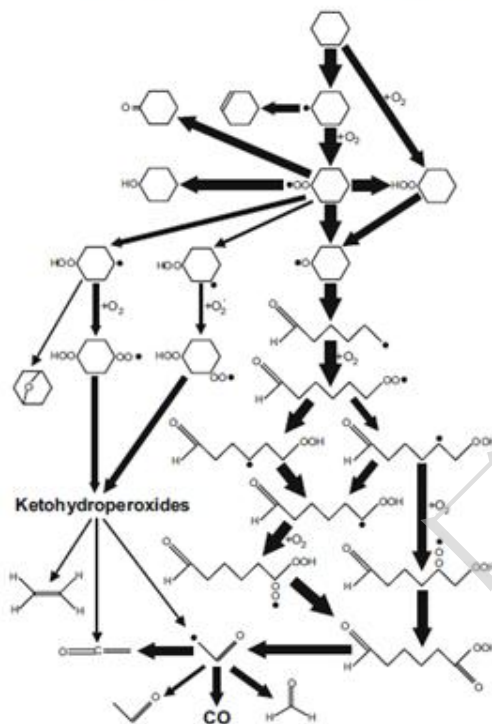
1. Fundamentals of Chemical Kinetics

I Complex Reactions and Analysis

Rate of Production Analysis (Reaction Flux)

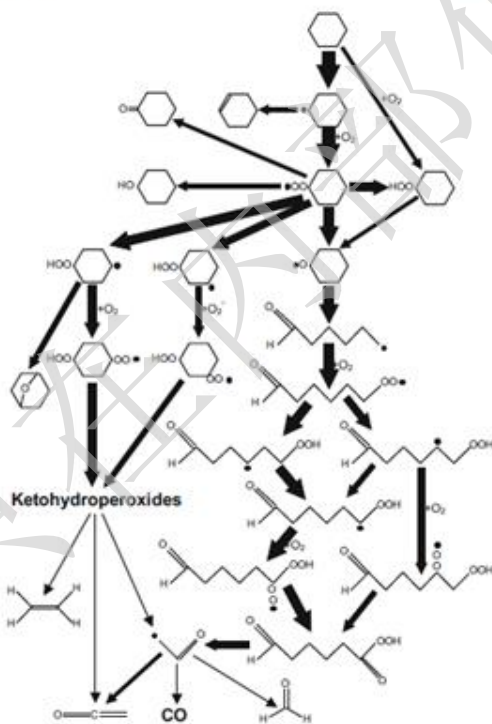
Production Contributions

$$\bar{C}_{ki}^p = \frac{\max(v_{ki}, 0)q_i}{\sum_{i=1}^I \max(v_{ki}, 0)q_i + \sum_{m=1}^M \frac{A_m}{V} \sum_{i=1}^{I^s} \max(v_{ki}^s, 0)q_i^s}$$



Destruction Contributions

$$\bar{C}_{ki}^d = \frac{\min(v_{ki}, 0)q_i}{\sum_{i=1}^I \min(v_{ki}, 0)q_i + \sum_{m=1}^M \frac{A_m}{V} \sum_{i=1}^{I^s} \min(v_{ki}^s, 0)q_i^s}$$



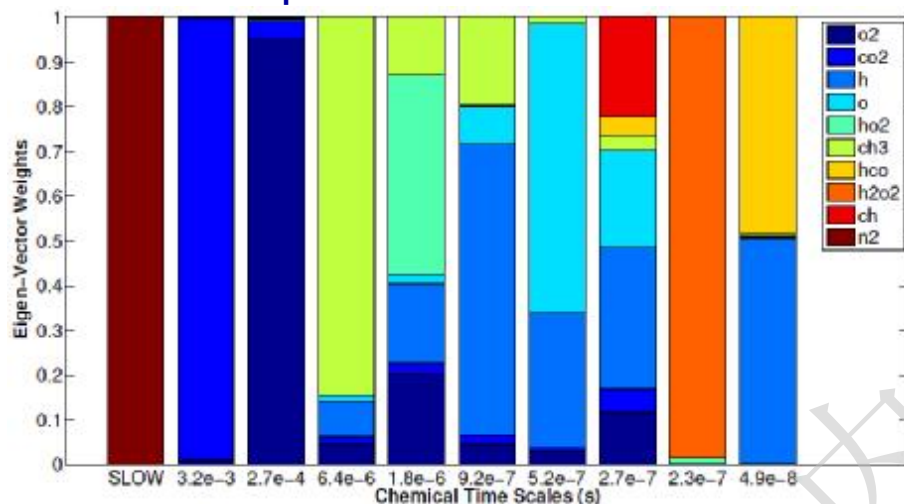
- n Show species conversion pathways
- n Analyze pathways leading to the consumption or production of a species
- n Analysis element fluxes
- n Analysis time-integrated element flux

- n Extremely useful in terms of investigating the influence of changing operating conditions on the dominant pathways

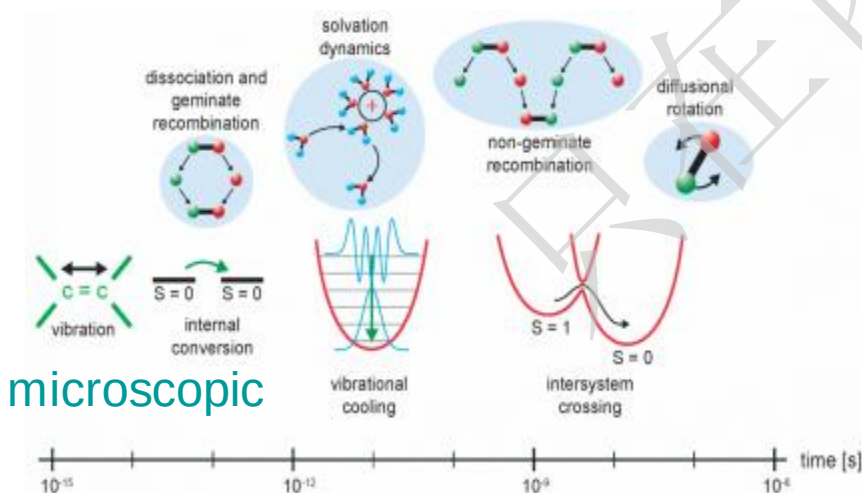
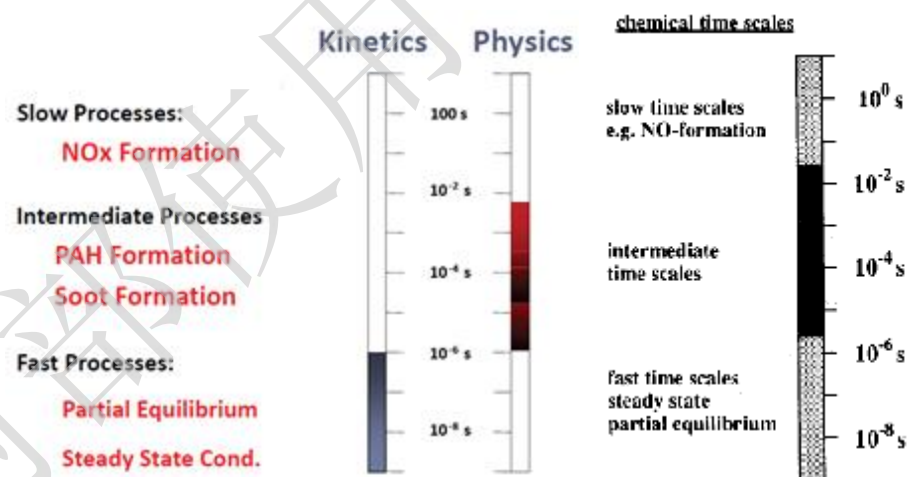
1. Fundamentals of Chemical Kinetics

I Time scales of complex reactions

Species time scales



Reactions time scales



- n The rate constants may differ from each other by several orders of magnitude
- n The concentrations of highly reactive intermediates may differ by orders of magnitude from stable reactants and products
- n The very fast time scales are usually responsible for equilibrium processes

2. Bimolecular Reaction Theory

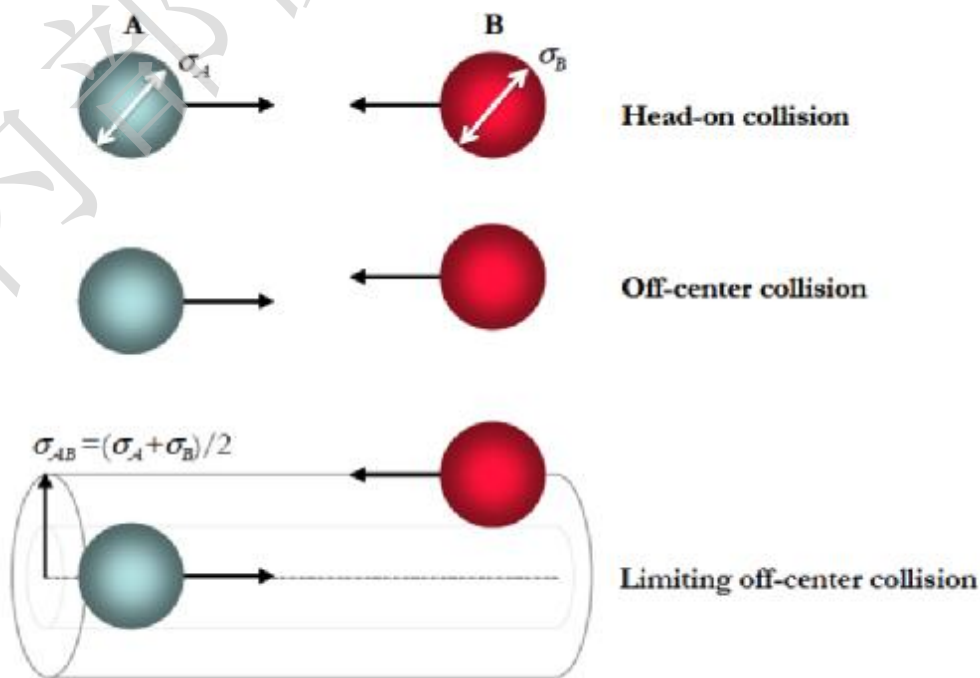
2.1. Collision Theory

2.2. Transition State Theory

2.1. Collision Theory

I Hard/Rigid Sphere Collision Theory

- n The rate coefficient of a bimolecular reaction ($A + B$) is the product of collision frequency Z_{AB} and reaction probability upon collision at a given temperature $\gamma(T)$ (determined by the fraction of molecules with combined energy greater than the activation energy) $k_{coll}[A][B] = Z_{AB}\gamma(T)$
- n The pre-exponential factor of Arrhenius expression ($k = A \exp(-\frac{E_a}{RT})$) is related to the collision frequency Z_{AB} , while the exponential term is related to the reaction probability $\gamma(T)$
- n The limiting off-center collision would have a spacing equal to $(\sigma_A + \sigma_B)/2$ between the two axes of molecular motions



Max Trautz (1916) and William C. M. Lewis (1918)

2.1. Collision Theory

I Hard/Rigid Sphere Collision Theory

- n Cross section within the cylindrical volume

$$s = \pi\sigma_{AB}^2 = \pi\left(\frac{\sigma_A + \sigma_B}{2}\right)^2$$

- n One A molecule, sweeping through a cylinder, the number of collisions per unit time

$$Z_B(s^{-1}) = \pi\sigma_{AB}^2 \bar{v}_A \frac{N_B}{V}$$

$\bar{v}_A$: Mean molecular velocity of A
 N_B/V : Number density of B

- n For N_A number of A molecules, collision rate

$$Z_{AB}(cm^{-3}s^{-1}) = \pi\sigma_{AB}^2 \bar{v}_{AB} \frac{N_A}{V} \frac{N_B}{V} \quad k_{coll}[A][B] = Z_{AB}\gamma(T)$$

- n Mean relative velocity of A and B given by the Maxwell velocity distribution function

$$\bar{v}_{AB} = \sqrt{\bar{v}_A^2 + \bar{v}_B^2} = \sqrt{\frac{8k_B T}{\pi m_{AB}}}$$

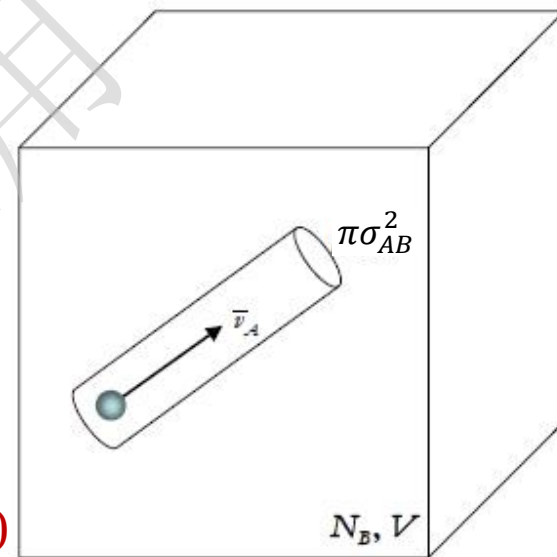
k_B : Boltzmann constant (1.3806×10^{-16})

m_{AB} : Reduced mass $m_{AB} = \frac{m_A + m_B}{m_A m_B}$

- n Rate coefficient at the collision limit ($\gamma(T) \equiv 1$)

$$k_{coll}(cm^3 mol^{-1} s^{-1}) = \pi\sigma_{AB}^2 \bar{v}_{AB} N_{avg} = \pi\sigma_{AB}^2 N_{avg} \sqrt{\frac{8k_B T}{\pi m_{A,B}}}$$

N_{avg} : Avogadro constant
(6.023×10^{23})



2.1. Collision Theory

I Hard/Rigid Sphere Collision Theory

Example: $\text{H}\cdot + \text{O}_2 \rightarrow \text{products}$

$$\sigma_{\text{H}\cdot} = 2.05 \text{ (\AA)} \text{ \& } \sigma_{\text{O}_2} = 3.46 \text{ (\AA)} \rightarrow \sigma_{\text{H}\cdot, \text{O}_2} = (\sigma_{\text{H}\cdot} + \sigma_{\text{O}_2})/2 = 2.755 \text{ (\AA)}$$

$$m_{A,B} = \frac{m_A + m_B}{m_A m_B} = \left(\frac{1 \times 32}{1 + 32} \right) \frac{1}{6.023 \times 10^{23}} = 1.61 \times 10^{-24}$$

$$k_{\text{coll}}(\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}) = \pi \times (1.61 \times 10^{-24})^2 \times 6.023 \times 10^{23} \sqrt{\frac{8 \times 1.3806 \times 10^{-16}}{\pi \times 1.61 \times 10^{-24}}} \times \sqrt{T}$$

$T \text{ (K)}$	$k_{\text{coll}} \text{ (cm}^3 \text{mol}^{-1} \text{s}^{-1})$
300	3.7×10^{14}
2000	6.7×10^{14}

At the same order of magnitude as the experimentally determined pre-exponential factor for this reaction (10^{14} - $10^{15} \text{ cm}^3 \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$)

2.1. Collision Theory

I Non-Rigid Sphere Collision Theory

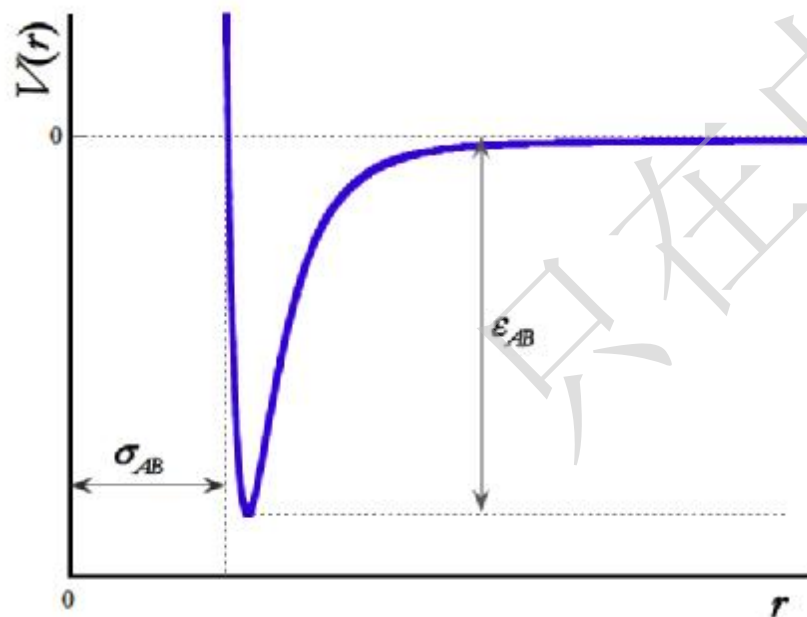
Lennard-Jones 12-6 potential (van der Waals interactions)

$$V_{vdW,AB}(r) = 4\varepsilon_{AB} \left[\left(\frac{\sigma_{AB}}{r} \right)^{12} - \left(\frac{\sigma_{AB}}{r} \right)^6 \right]$$

ε_{AB} : Collision well depth

σ_{AB} : Collision diameter

r : Distance of intermolecular separation



Reflection angle

$$\chi = \pi - 2b \int_{r_m}^{\infty} \frac{dr}{r^2} \sqrt{1 - \frac{2V_{vdW,AB}(r)}{m_{AB}v_r^2} - \frac{b^2}{r^2}}$$

Collision cross section

$$S = 2\pi \int_0^{\infty} (1 - \cos \chi) b db$$

Collision integral

$$\Omega^* = \frac{S}{\pi \sigma_{AB}^2}$$

$$k_{coll} (cm^3 mol^{-1} s^{-1}) = \sqrt{\frac{8k_B T}{\pi m_{AB}}} \pi \sigma_{AB}^2 \Omega^* N_{avg}$$

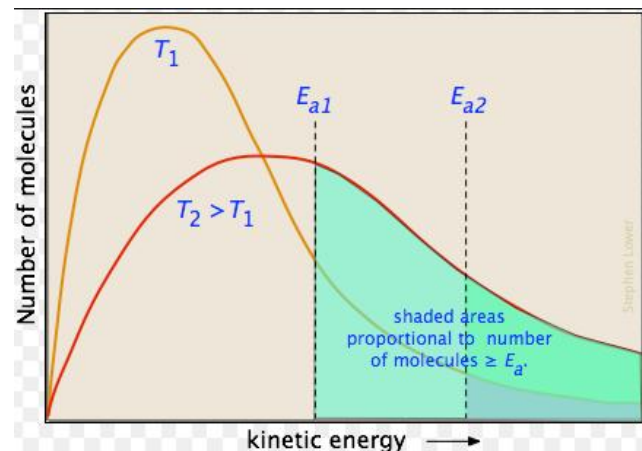
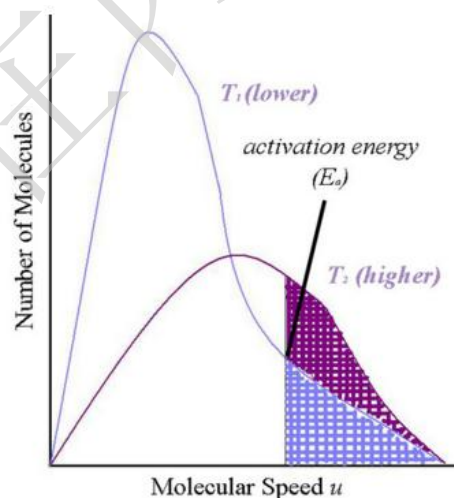
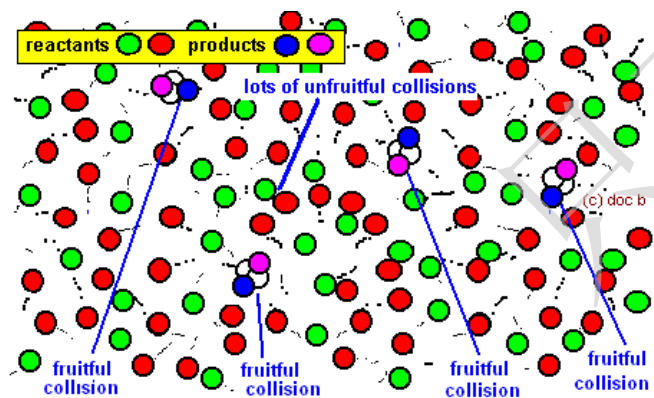
- Non-rigid sphere collision introduces the reduced collision integral, accounts for the effect of **intermolecular forces** on the collision cross section

Enough or not?

2.1. Collision Theory

I Summary

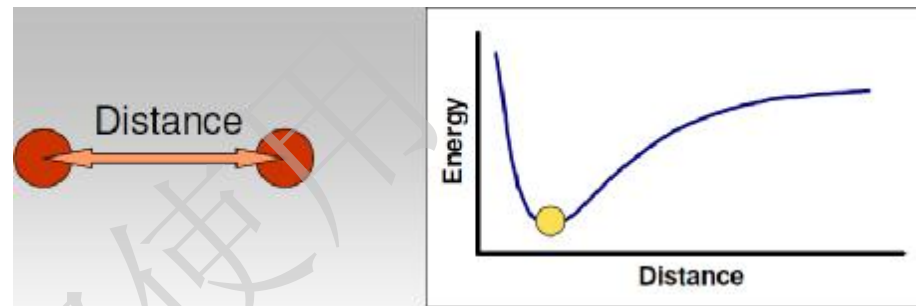
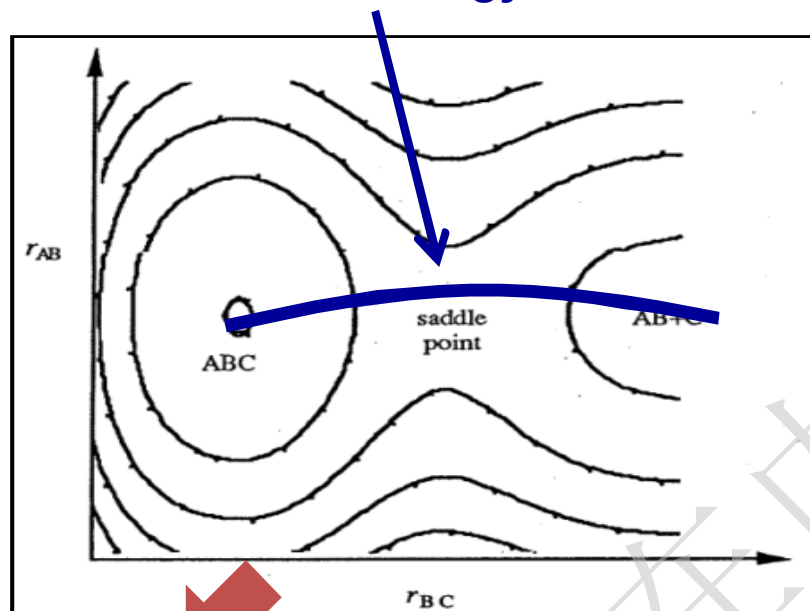
- n The collision theory sets the upper boundary for the bimolecular reaction rate coefficient, but it is still a simple and rough theory
- n The upper limits of rate constants are at 10^{13} - 10^{14} $\text{cm}^3\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$ for bimolecular reactions without the H atom as one of the reactants
- n The upper limits of rate coefficients are at 10^{14} - 10^{15} $\text{cm}^3\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$ for reactions with the H atom



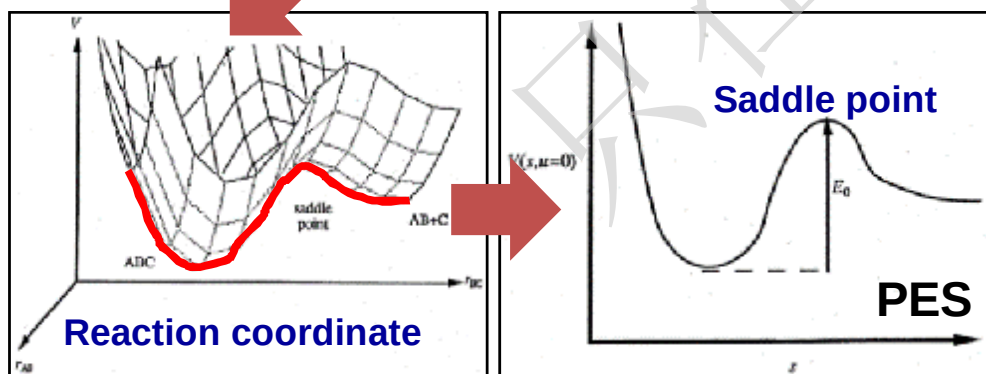
2.2. Transition State Theory

I Potential Energy Surface (PES)

Minimum Energy Path

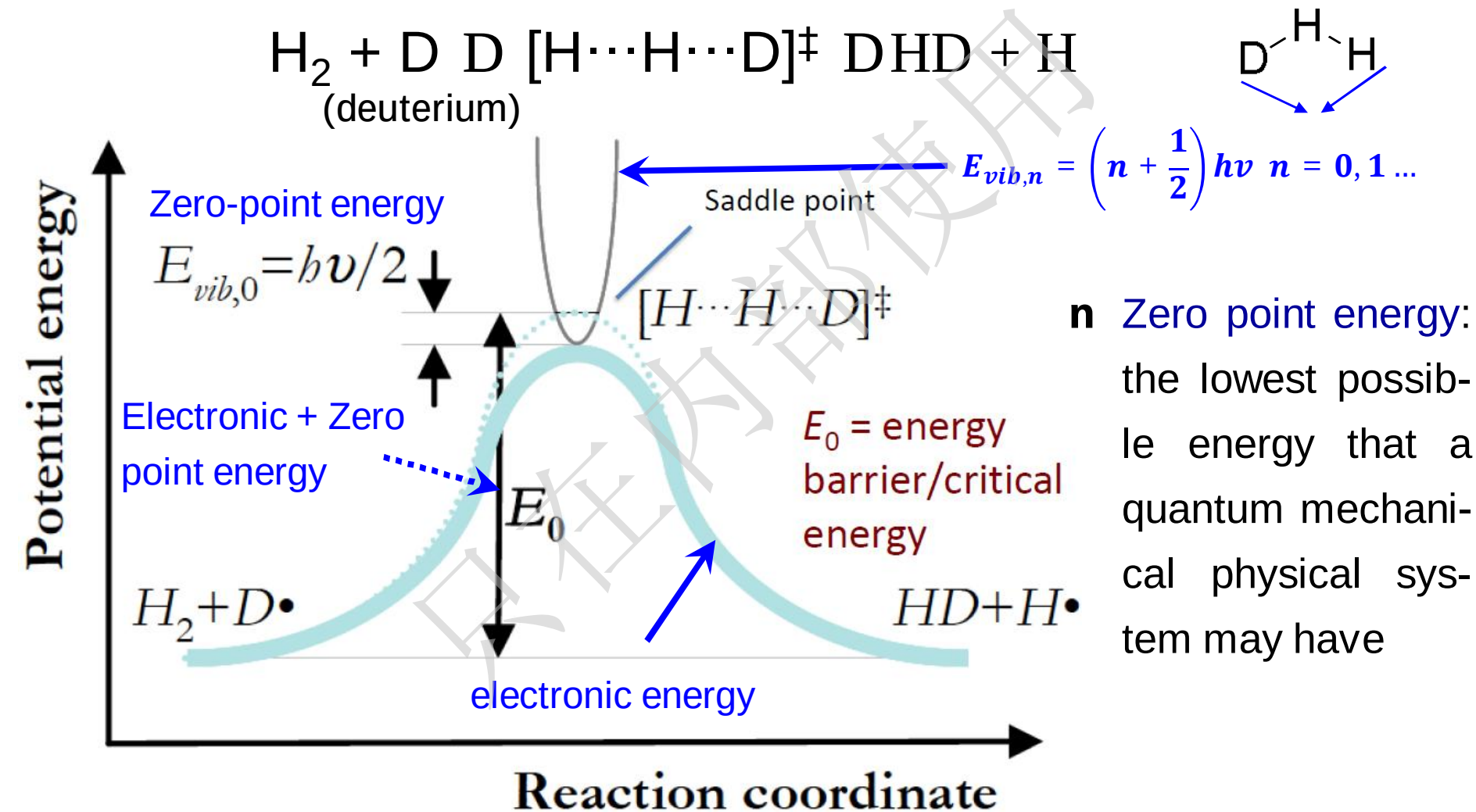


- n PES is the energy of a molecule as a function of the locations of its nuclei
- n The **minimum electronic energy path** is along the bottom of the valley, this path is the **reaction coordinate**
- n The highest point on the reaction coordinate is the **saddle point**, is the minimum potential energy that the complex must overcome to proceed from reactants to products



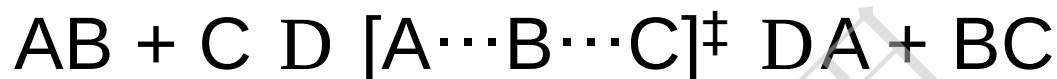
2.2. Transition State Theory

I Potential Energy Surface



2.2. Transition State Theory

I Theory Principles

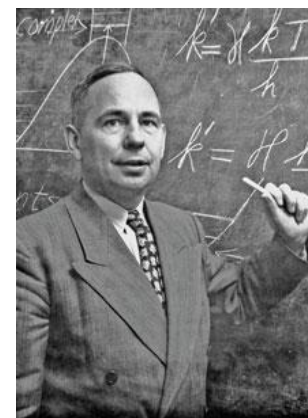
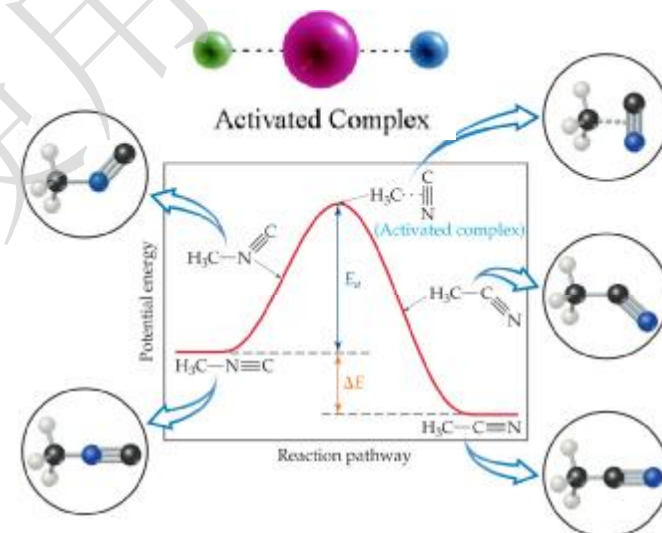


Theory:

- n The **transition state** is the highest potential energy point on the reaction coordinate
- n The molecular structure at the transition state is called the **activated complex**

Assumptions:

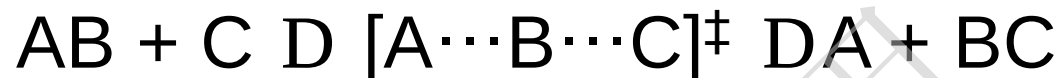
- n Population of the activated complex is in equilibrium with the reactants
- n Activated complex can be treated as a distinct chemical species
- n Activated complex decompose rapidly, on the order of a vibrational period (10^{-13} - 10^{-14} s or 10-100 fs)



Henry Eyring (1935)

2.2. Transition State Theory

I Partition Function Formulation of Rate Constant



$$k_{TST} = \frac{k_B T}{h} \frac{Q^{\ddagger}}{\prod Q_R} \exp(-E^{\ddagger}/k_B T)$$

- n h - Planck's constant (6.626×10^{-34} J·s)
- n Q - **Partition function**: The distribution of particles in each quantum state, a function of temperature and volume ($Q^{\ddagger}$ and Q_R for activated complex and reactants)
- n Properties of the activated complex, reactants and products can be determined from electronic structure calculations (quantum chemistry calculations)

2.2. Transition State Theory

I Thermodynamic Formulation of TST

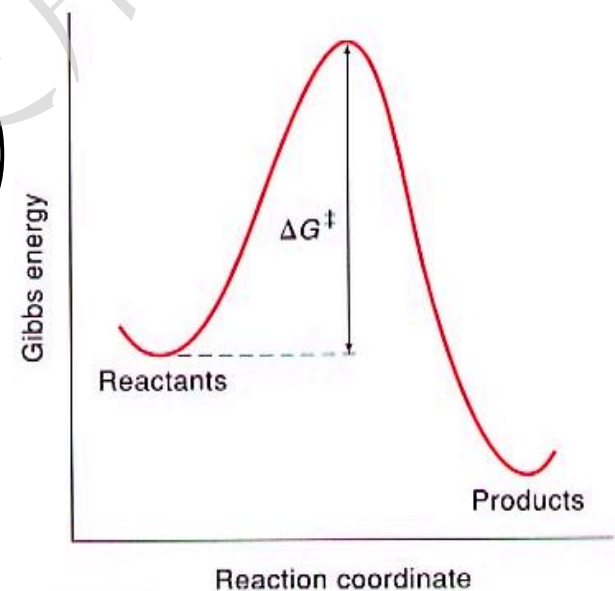
$$K_{TST} = \frac{[A \cdots B \cdots C]^{\ddagger}}{[AB][C]} = (RT) \exp\left(-\frac{\Delta G^{\ddagger}}{RT}\right)$$

$$k_{TST} = \frac{k_B T}{h} \frac{Q^{\ddagger}}{\prod Q_R} \exp\left(-\frac{E^{\ddagger}}{k_B T}\right) = \frac{k_B T}{h} (RT) \exp\left(-\frac{\Delta G^{\ddagger}}{RT}\right)$$

$$k_{TST} = \frac{k_B T}{h} (RT) \exp\left(\frac{\Delta S^{\ddagger}}{R}\right) \exp\left(-\frac{\Delta H^{\ddagger}}{RT}\right)$$

$$E_a = 2RT + \Delta H^{\ddagger} \quad A = \frac{k_B T}{h} (RT) e^2 \exp\left(\frac{\Delta S^{\ddagger}}{R}\right)$$

$$\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$$



- n The activation energy is related to the energy barrier E_a
- n The pre-exponential factor A is related to the entropy change

2.2. Transition State Theory

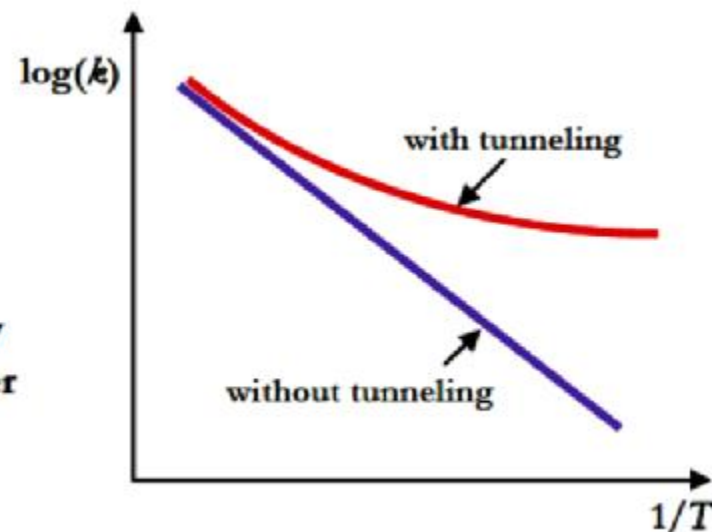
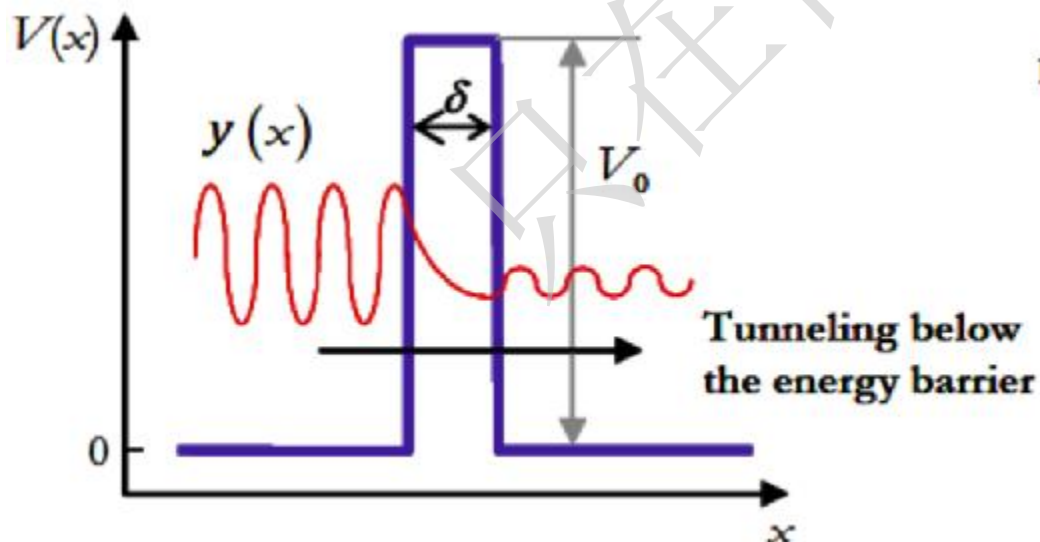
I Quantum Tunneling Effect

- n There is a finite probability for a molecule to tunnel through the potential energy barrier based on the Schrödinger equation
- n The energy barrier of a reaction may be viewed as an imperfect rectangular potential barrier, of finite width, has a non-zero probability for the reaction to undergo tunneling

n

me

At which T the tunneling effect will become important?



2.2. Transition State Theory

I Summary

- n TST is the foundation of modern reaction rate theory
- n TST gives a more detailed account of the reaction process that considers the influence of potential energy barrier and the nature of the activated complex
- n TST predicts that the rate constant of a bimolecular reaction is dependent on temperature only, if there is only one energy barrier separating the reactants and products
- n Reaction types include metathesis reaction, H-atom abstraction reactions and reactions that exchange atoms or groups between two reactants
- n TST is extremely useful for extrapolation of rate constant data from a limited temperature range to the entire range of combustion temperatures

2.2. Transition State Theory

I Summary

n Expressions of rate constant

**Arrhenius Expression
(Empirical)**

$$k = A \exp\left(-\frac{E_a}{RT}\right)$$

**Collision Theory
(Simple theory)**

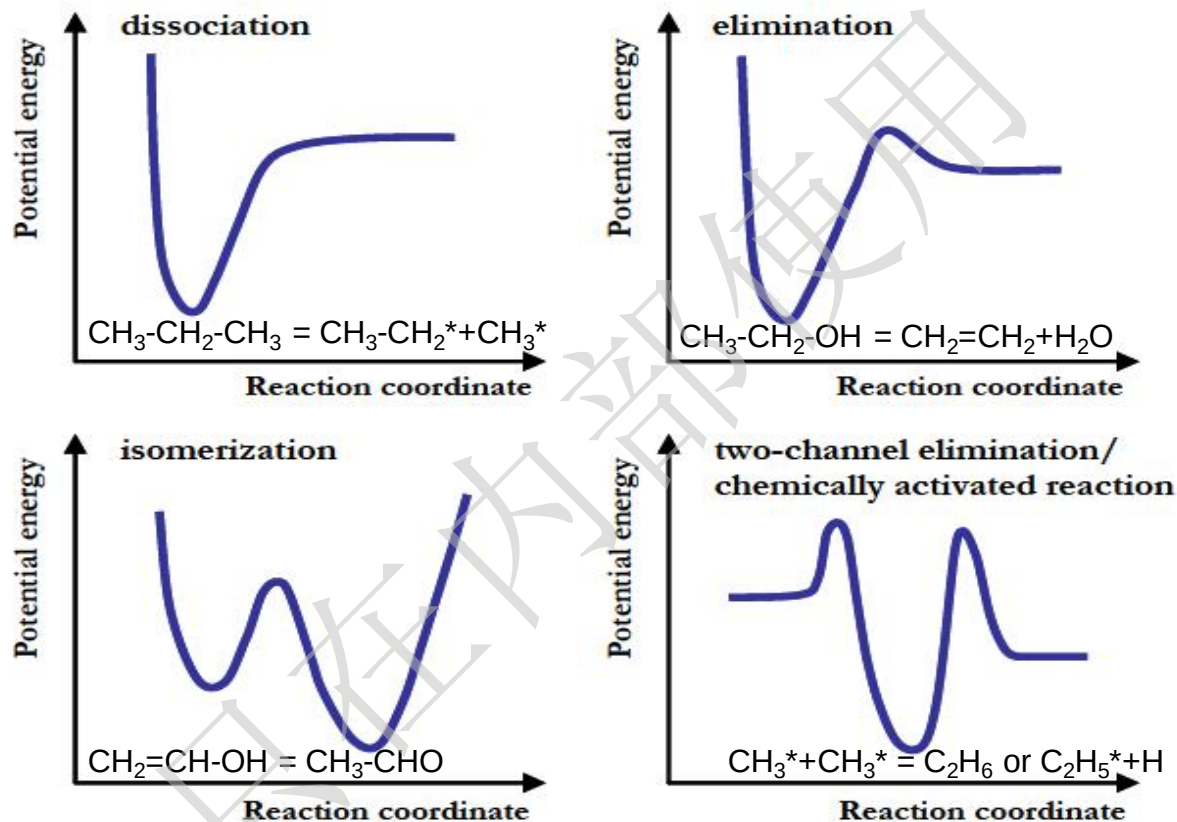
$$k_{coll} = \pi \sigma_{AB}^2 N_{avg} \sqrt{\frac{8k_B T}{\pi m_{A,B}}} \gamma(T)$$

**Transition State Theory
(Modern rate theory)**

$$k_{TST} = \frac{k_B T}{h} (RT) \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right)$$

3. Unimolecular Reaction Theory

I Pressure-Dependent Reaction

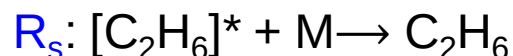
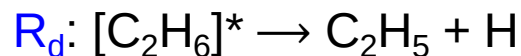
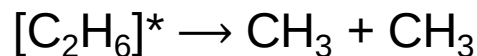
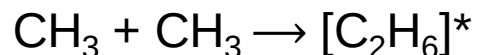


- n Collision activation process: thermal activation, chemically activation
- n Unimolecular reaction theory is also applicable to a large number of bimolecular reactions (e.g. chemically activated reactions)

Why?

3. Unimolecular Reaction Theory

I Chemically Activated Reaction



Chemically activation

Deactivation

Dissociation

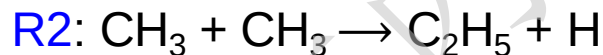
Stabilization



Boltzmann
distribution



Net observables



- n Population of the activated complex $[\text{C}_2\text{H}_6]^*$ is dependent on pressure
- n Rate of R_s increases with the increasing third body concentration
- n Rate constants of $R1$ and $R2$ are pressure dependent

3.1. Lindemann Mechanism

I Nature of unimolecular reactions

Lindemann mechanism (1922)

R1f: $A + M \rightarrow A^* + M$ collision activation

R1b: $A^* + M \rightarrow A + M$ collision deactivation

R2: $A^* \rightarrow \text{products}$ spontaneous reaction

n A^* is the reactant capable of overcoming the *potential energy barrier* to form the products.

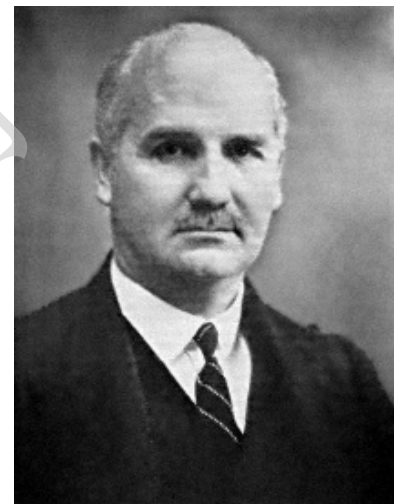
n For a simple bond dissociation reaction, this energy barrier is equal to the bond dissociation energy (BDE)

$$\frac{d[A^*]}{dt} = k_{1f}[A][M] - (k_{1b}[M] + k_2)[A^*]$$

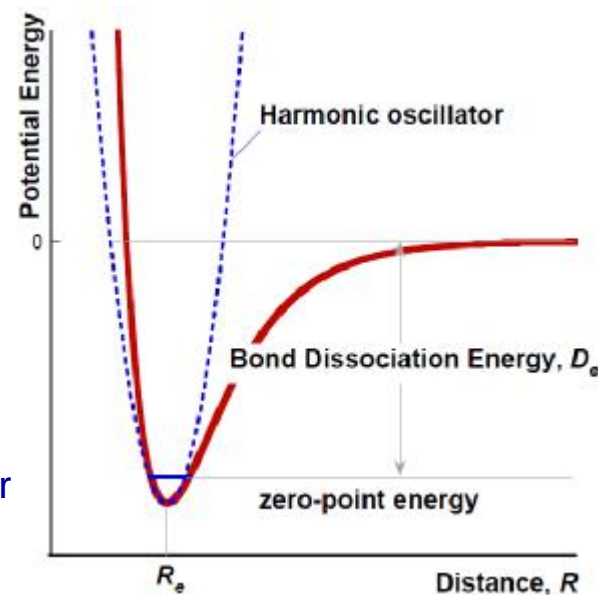
$$\frac{d[A^*]_{QSS}}{dt} = k_{1f}[A][M] - (k_{1b}[M] + k_2)[A^*]_{QSS} = 0$$

$$-\frac{d[A]}{dt} = \left\{ \frac{1}{k_2 k_{1f}/k_{1b}} + \frac{1}{k_{1f}[M]} \right\}^{-1} [A]$$

$$k_{uni} = k_2 \frac{k_{1f}}{k_{1b}} \left/ \left\{ 1 + \frac{k_2}{k_{1b}[M]} \right\} \right. \quad \text{Rate constant of unimolecular decomposition reaction}$$



Frederick A. Lindemann



3.1. Lindemann Mechanism

I Nature of unimolecular reactions

- n Low-P limit: $P \rightarrow 0$, $[M] = P/RT \rightarrow 0$

$$k_{uni,0} = k_{1f}[M]$$

$$k_{uni} = k_2 \frac{k_{1f}}{k_{1b}} \left/ \left\{ 1 + \frac{k_2}{k_{1b}[M]} \right\} \right.$$

At low-P limit, the rate constant of a unimolecular reaction is proportional to pressure

- n High-P limit: $P \rightarrow \infty$, $[M] \rightarrow \infty$

$$k_{uni,\infty} = \frac{k_2 k_{1f}}{k_{1b}}$$

At high-P limit, the rate constant of a unimolecular reaction is a constant value

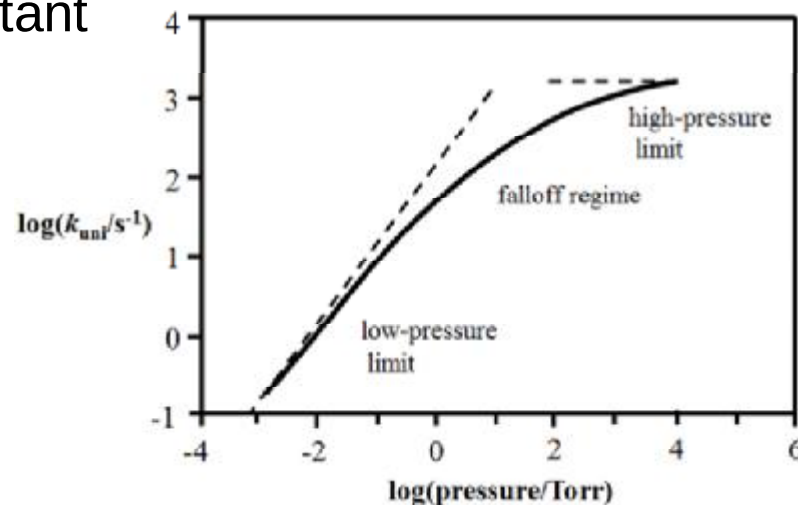
- n Overall pressure-dependent rate constant

$$\frac{1}{k_{uni}} = \frac{1}{k_{uni,0}} + \frac{1}{k_{uni,\infty}}$$



- n For bimolecular combination reaction

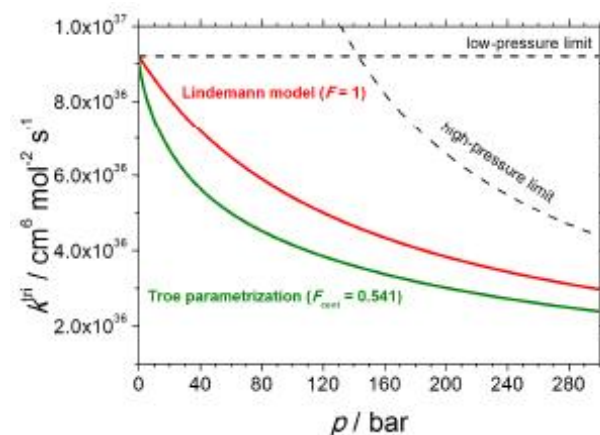
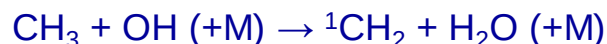
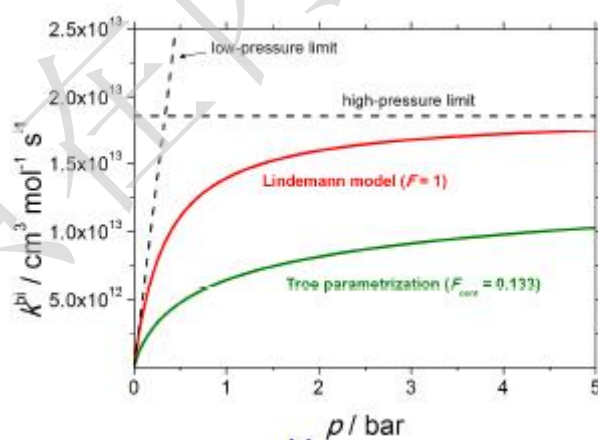
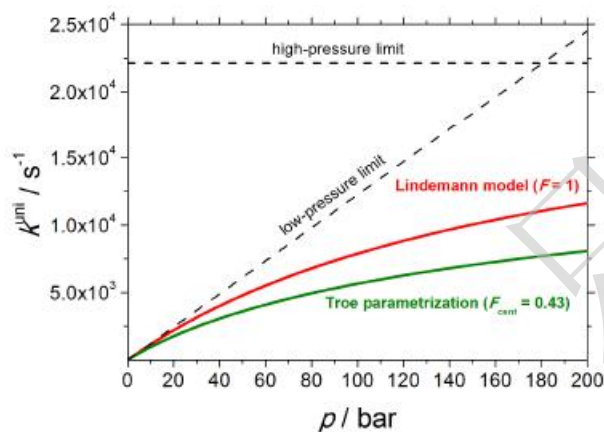
$$k_{bi} = \frac{k_{uni}}{K_c}$$



3.1. Lindemann Mechanism

I Fall-off of Rate Coefficient

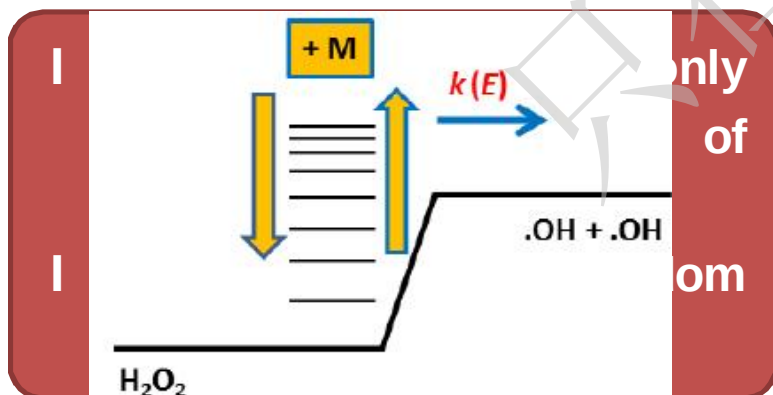
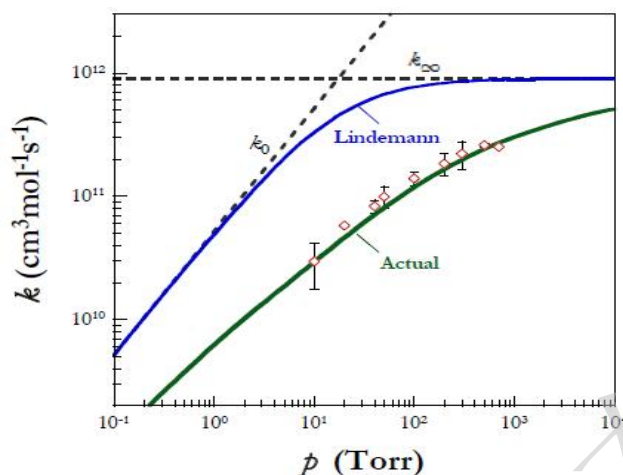
- As P decreases, k_{uni} will drop from the high- P limit to the low- P limit, called as the fall-off behavior
- The rate constants for small molecules (e.g. diatomics) tend to be close to the low- P limit at atmospheric pressure
- The rate constants for large molecules tend to be close to the high- P limit at atmospheric pressure



3.1. Lindemann Mechanism

I Hinshelwood-Lindemann Mechanism (1926)

- n Lindemann mechanism fails to predict the rate coefficient in a quantitative fashion



- n Thermally activated molecule can assume different rates of dissociation depending on its energy, with an **energy-dependent rate constant** $k(E)$ $A^*(E) \rightarrow \text{products}$
- n This rate constant is basically a frequency factor of a molecule with vibrational energy above the zero-point to undergo chemical transformation
- n Requires that k_{uni} be treated by considering **$k(E)$ averaged over the populations at all energy levels above the energy barrier**

$$k_{\text{uni}} \propto \frac{\sum_i k(E_i) f(E_i)}{\sum_i f(E_i)} \quad \mathbf{f(E_i)}: \text{population at the } i^{\text{th}} \text{ energy level}$$

3.2. RRKM Theory

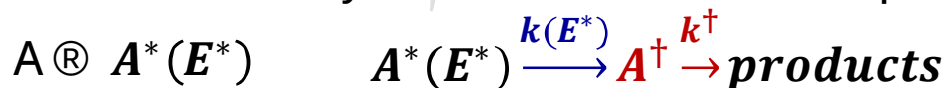
I Theory Principles

Rice–Ramsperger–Kassel (RRK) theory (1928)

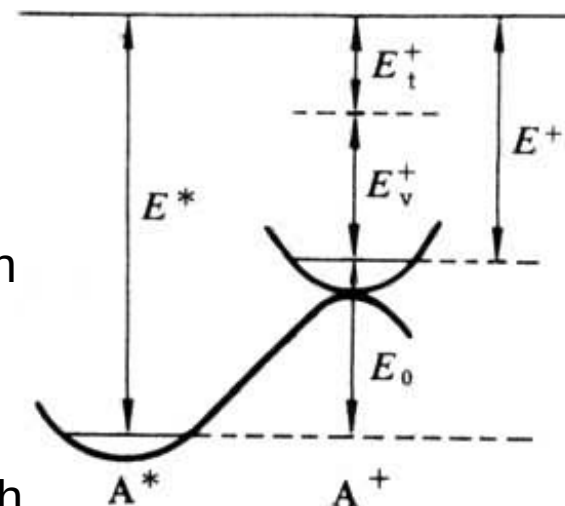
- n Developed based on the Hinshelwood-Lindemann mechanism
- n Assumes that energy can flow freely from one vibrational mode to another within the molecule
- n An incomplete theory due to the following limitations: a) arbitrarily chooses vibrational degrees of freedom; b) uses same frequencies for the vibrational modes of excited reactant and activated complex

RRK-Marcus (RRKM) theory (1952)

- n Ergodicity assumption: the rapid randomization of energy throughout vibrational degrees of freedom
- n The existence of a **critical geometry** for the transition from a vibrationally excited molecule to the product

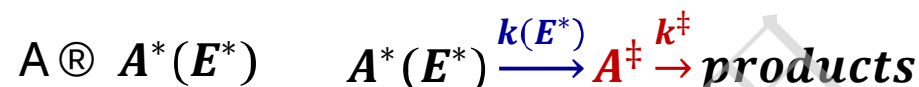


- n Connects Hinshelwood-Lindemann mechanism with statistical and TST theories



3.2. RRKM Theory

I Formulation of Rate Constant



- n $k^\ddagger$ is of the order of a vibrational frequency and $k^\ddagger \geq k(E^*)$, thus it is key to determine $k(E^*)$

$$k(E^*) = k^\ddagger \frac{[A^\ddagger]}{[A^*(E^*)]}$$

- n By considering the equilibrium and participating the density of states of the activated complex:

$$k_{RRKM}(E) = \frac{\int_0^{E-E_0} \rho^\ddagger(E^\ddagger) dE^\ddagger}{h\rho(E)} = \frac{N(E^\ddagger)}{h\rho(E)}$$

The sum of states of the activated complex

The density of energy states of the reactant

- n Account for the rotational energy variations:

$$k_{RRKM}(E) = \frac{q_r^\ddagger N(E^\ddagger)}{q_r h \rho(E)}$$

- n Consider the reaction path degeneracy: $k_{RRKM}(E) = l_a \frac{q_r^\ddagger N(E^\ddagger)}{q_r h \rho(E)}$

3.2. RRKM Theory

I History of Unimolecular Reaction Theory

N. Slater, 1955

Slater theory

Classic mechanics

Rely on heavy mathematical treatments, poor accuracy

R. A. Marcus (1952) introduced TS theory developed by H. Eyring (1935) and finalized the whole theory

RRKM theory

statistics

RRK theory

O. K. Rice and H. C. Ramsperger (1927) and L. S. Kassel (1928)

Hinshelwood-Lindemann mechanism

Developed by C. Hinshelwood (1926) considering the role of vibrational degrees of freedom

Lindemann mechanism

Discovered by F. Lindemann in 1922

Foundation of modern reaction kinetics

...

3.3. Parameterization for Modeling

I Chemkin format and Troe formula

```

2OH(+M)<=>H2O2(+M) 2.148E+05 2.3219 -1712.31
LOW / 1.960E+015 -0.2033311 -2174.6881 /
TROE / 4.300E-01 1.0E-30 1.0E+30 1.0E+30 /
H2O/5/ AR/0.67/ O2/0.8/ HE/0.43/ H2O2/5.13/ H2/2.47/
    
```

→ Third body

Collision

Efficiency

Parameters

Troe parameters

$$k_0 = A_0 T^{\beta_0} \exp(-E_0/R_c T)$$

$$k_\infty = A_\infty T^{\beta_\infty} \exp(-E_\infty/R_c T)$$

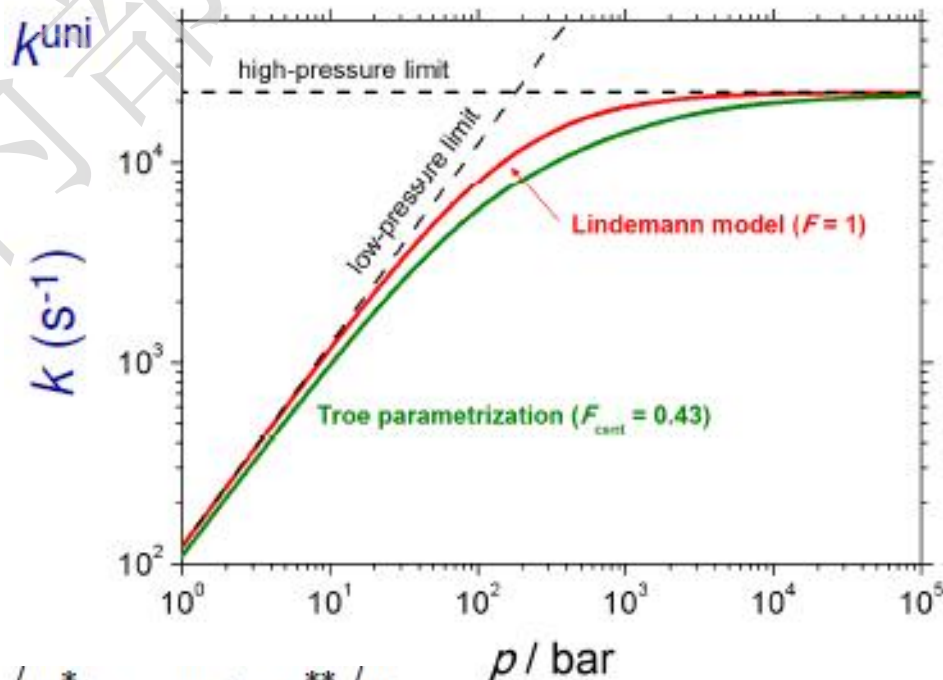
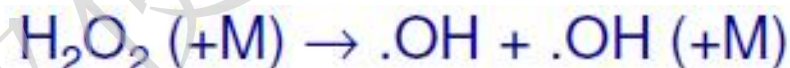
$$k = k_\infty \left(\frac{P_r}{1 + P_r} \right) F \quad P_r = \frac{k_0 [M]}{k_\infty}$$

$$\log F = \left[1 + \left[\frac{\log P_r + c}{n - d(\log P_r + c)} \right]^2 \right]^{-1} \log F_{\text{cent}}$$

$$c = -0.4 - 0.67 \log F_{\text{cent}} \quad d = 0.14$$

$$n = 0.75 - 1.27 \log F_{\text{cent}}$$

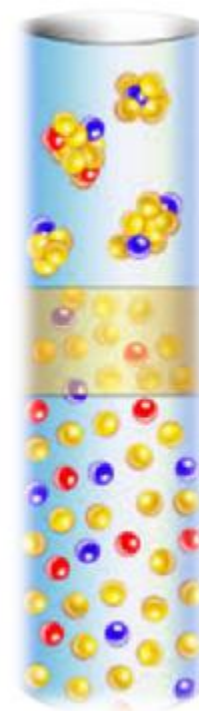
$$F_{\text{cent}} = (1 - \alpha) \exp(-T/T^{***}) + \alpha \exp(-T/T^*) + \exp(-T^{**}/T)$$



3.3. Parameterization for Modeling

I Third body collision efficiency

- n **M**: Third body with collision efficiency parameters
- n Any species present in the mixture (molecule, radical, atom ...)
- n Good colliders: 1) species with similar energy levels to those of the excited species; 2) large molecules with many energy levels
- n Examples: H₂O can store energy in many rotational and vibrational modes; N₂ can store energy in translational, rotational and vibrational modes
- n Poor collider: e.g. noble gases: no rotational or vibrational energy levels, only the translational mode can be excited
- n Calculation of the effective concentration of M $[M] = \sum_i m_i [Y_i]$



2OH(+M)<=>H2O2(+M)	2.148E+05	2.3219	-1712.31
LOW /	1.960E+015	-0.2033311	-2174.6881 /
TROE /	4.300E-01	1.0E-30	1.0E+30 1.0E+30 /
H2O/5/ AR/0.67/ O2/0.8/ HE/0.43/ H2O2/5.13/ H2/2.47/CO/1.87/CO2/1.07			

$$[M] = 5[H_2O] + 5.13[H_2O_2] + 0.8[O_2] + 2.47[H_2] + 1.87[CO] + 1.07[CO_2] + 0.67[Ar] + 0.43[He] + [\text{all others}]$$

4. Recommended Literatures

1. Chapters 20 and 21, *Atkins' Physical Chemistry*, 9th edition, edited by Peter Atkins and Julio de Paula, December 18, 2009. *For ones who want to know more about the basic concepts of reaction kinetics.*
2. Chapter 2, *Comprehensive chemical kinetics*, edited by M.J. Pilling, November 1997. *For ones who want to know more about elemental reaction and bimolecular reaction rate theory and mechanism analysis methods.*
3. Chapter 4, *Comprehensive chemical kinetics*, edited by R.W. Carr, September 2007. *For ones who want to know more about unimolecular reaction theory and master equation.*
4. Chapters 2-6, *Unimolecular reaction dynamics: theory and experiments*. edited by T. Baer and W.L. Hase, 1996. *For ones who want to know more about unimolecular reaction theory and some knowledge about reaction dynamics.*

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